

Supplementary Online Content

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eMethods

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This supplementary material was provided by the authors to give readers additional information about their work.

Association of the Extent of ischemic Core and Penumbra with the Benefits of Endovascular Thrombectomy: Analysis of the SELECT2 Trial

Supplementary materials:

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eMethods

Analysis comparing explanatory capabilities of various quantitative ischemic injury estimates:

In order to provide a head-to-head comparison of explanatory capabilities of quantitative ischemic injury estimates on CT Perfusion (using rCBF <30% threshold), Non-contrast CT (Manually delineated CT hypodensity volume) and considering both images together (composite core volume), information criteria and area-under-the-curve (for dichotomous functional independence and independent ambulation) were reported for models including age, NIHSS-score at presentation, time from last known well to randomization, and one imaging parameter (CT-ASPECTS, CT-hypodensity volume, CT-perfusion/MR-diffusion core volume or composite core volume) to compare model fit for functional outcomes. Lower information criteria values and higher area-under-the-curve (AUC) values were considered better. By convention, Bayesian information criterion (BIC) differences were categorized as 0-2 weak, 2-6 positive, 6-10 strong and >10 very strong evidence of model superiority.¹

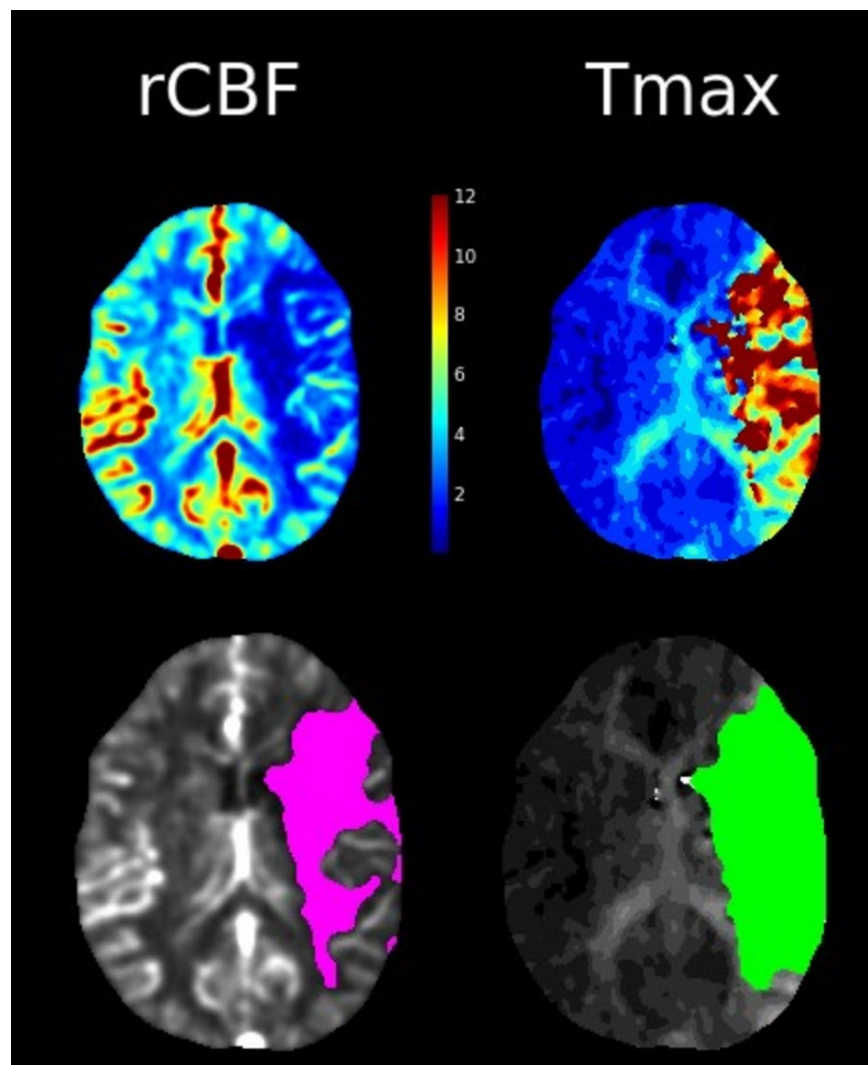
Reference:

1. Raftery AE. Bayesian model selection in social research. *Sociol Methodol*. Published online 1995:111-163.

eFigures

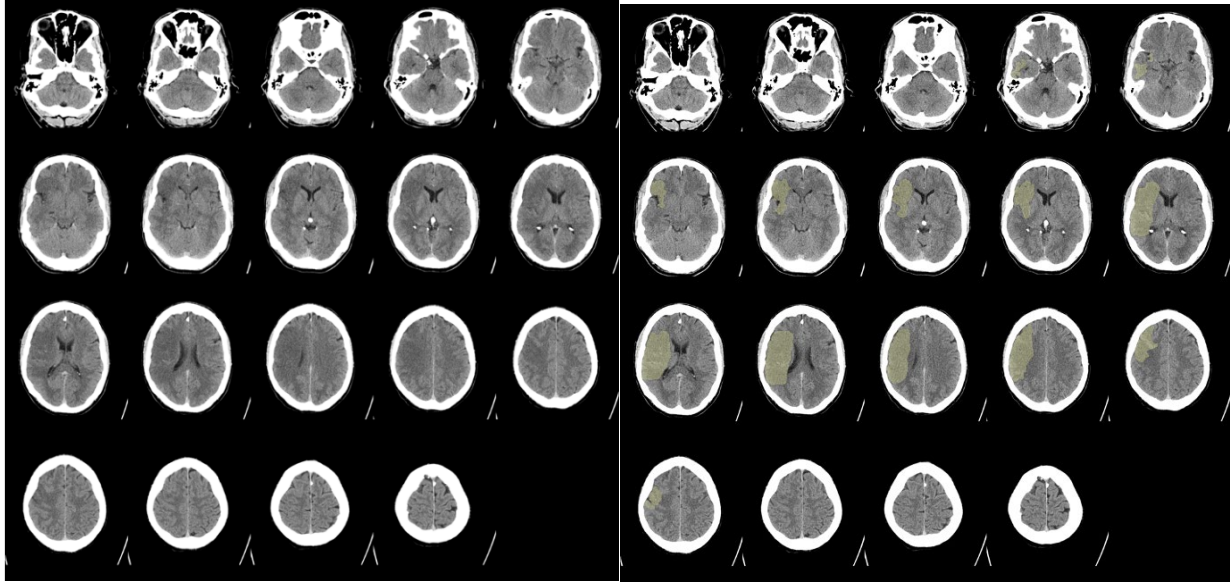
eFigure 1. Illustrative CT Perfusion imaging in a patient with occlusion of the M1 segment of left middle cerebral artery.

The overall rCBF and Tmax maps are provided in the first series, whereas the second series provides an illustration for maps using rCBF<30% and Tmax>6s thresholds. The region with severely reduced relative cerebral blood flow (rCBF) <30% (pink region on the left) is estimated to be irreversibly injured due to ischemia. The region with time to maximum intensity (Tmax) >6 s (green region on the right) represents delayed perfusion and thus brain area at the risk of infarction in the absence of restored blood flow (reperfusion). The ratio of tissue volume with 1) Tmax >6s and 2) rCBF <30% is termed the mismatch ratio. The volumetric difference between the two regions is termed the mismatch volume and estimates the volume of ischemic penumbra (brain tissue that is at risk of infarction but can be salvaged if blood flow to the area is rapidly restored). For this illustrative case, the rCBF <30% volume was 94ml and Tmax >6s volume was 131ml. Thus, mismatch ratio was 1.4 and mismatch volume was 37 ml.

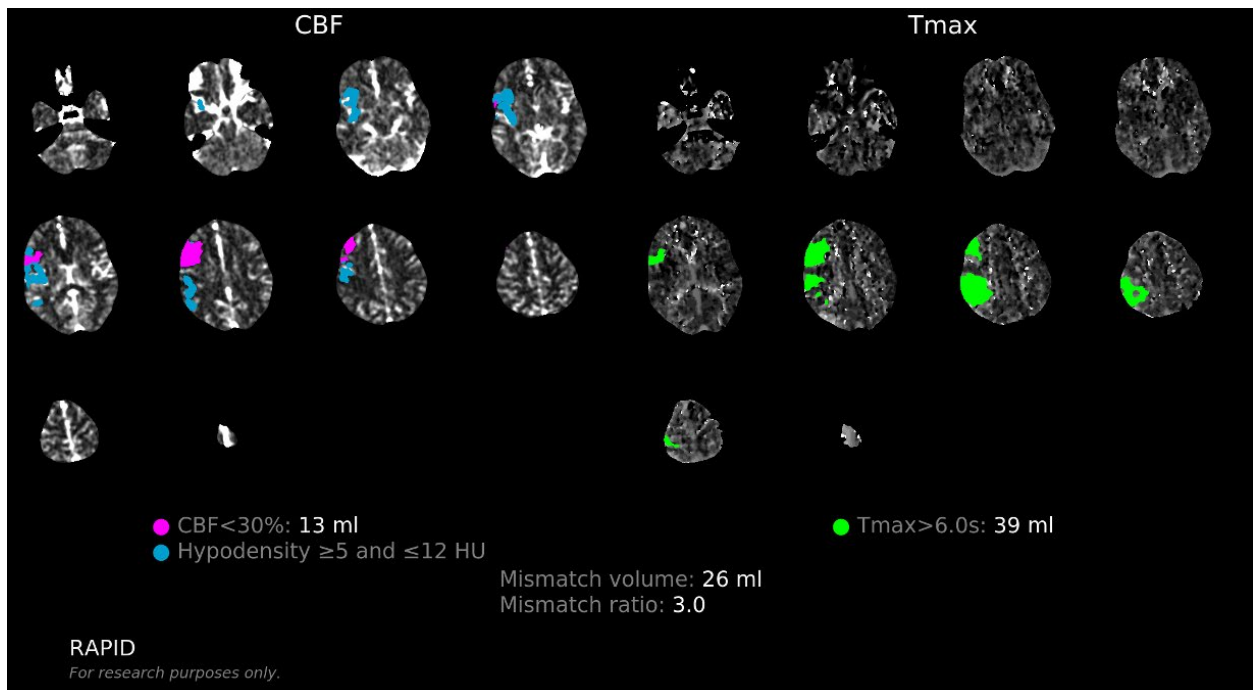


eFigure 2. Illustrative case of core estimation on non-contrast CT and CT-perfusion at baseline and follow-up infarct volume.

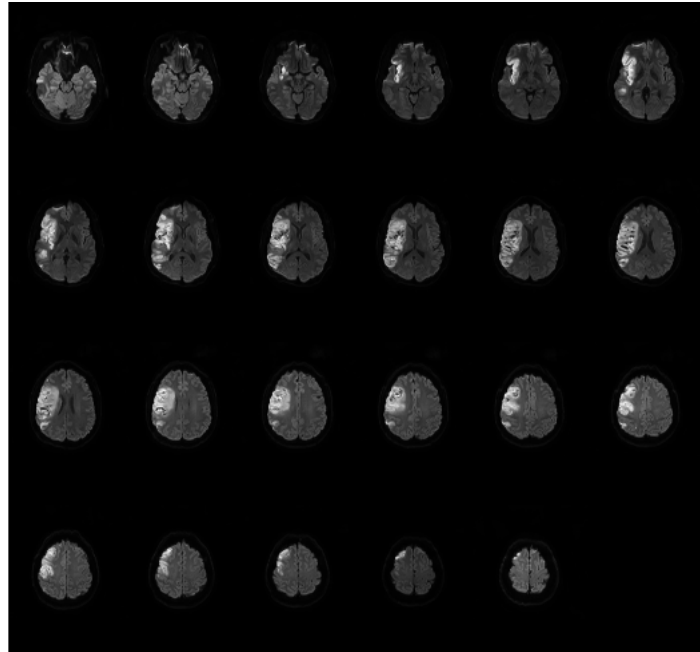
A) Non-contrast CT at baseline with significant hypodensity in right fronto-parieto-temporal lobe without and with manual delineation of hypodensity area



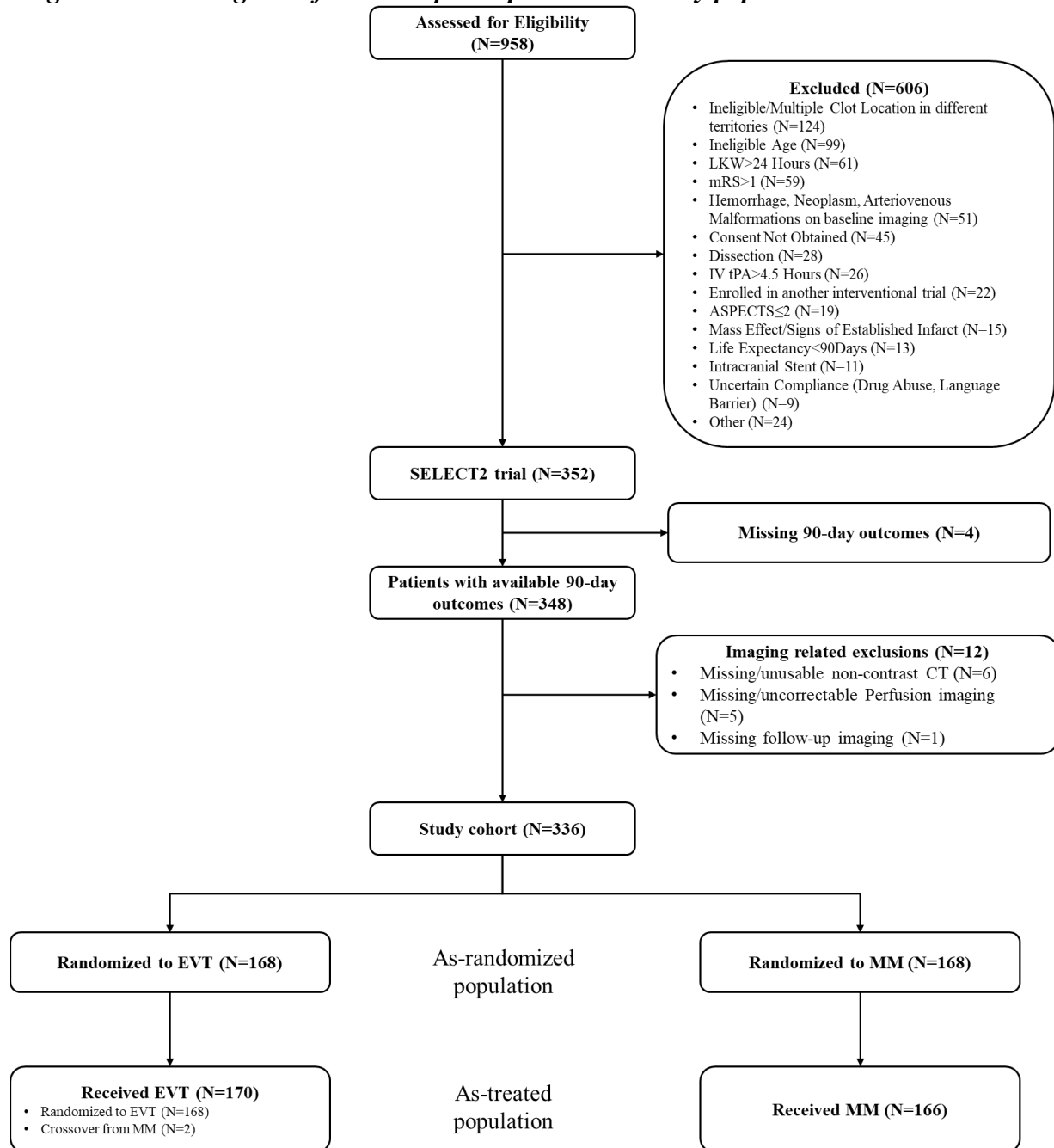
B) The same case with CT-perfusion demonstrating ischemic core based on $rCBF < 30\%$ volume. Note hypodensity feature identifying ischemic area outside of ischemic core region based on CTP, likely due to partial reperfusion / recruitment of collaterals.



C) Follow-up DWI demonstrating infarcted brain tissue in the region corresponding to the CT hypodensity

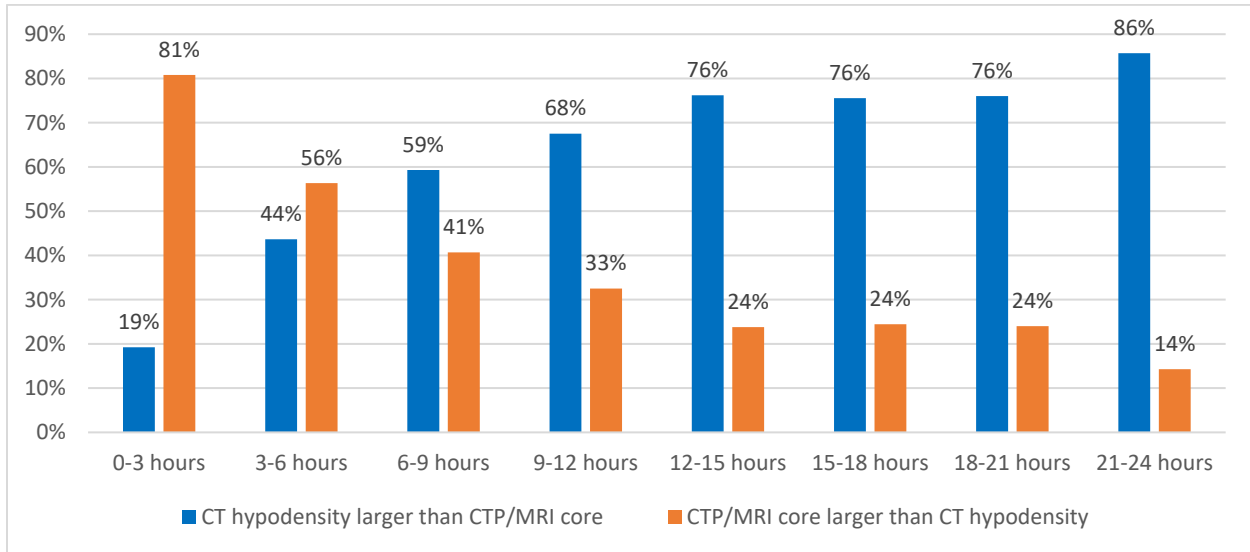


eFigure 3. Flow diagram of included participants in the study population



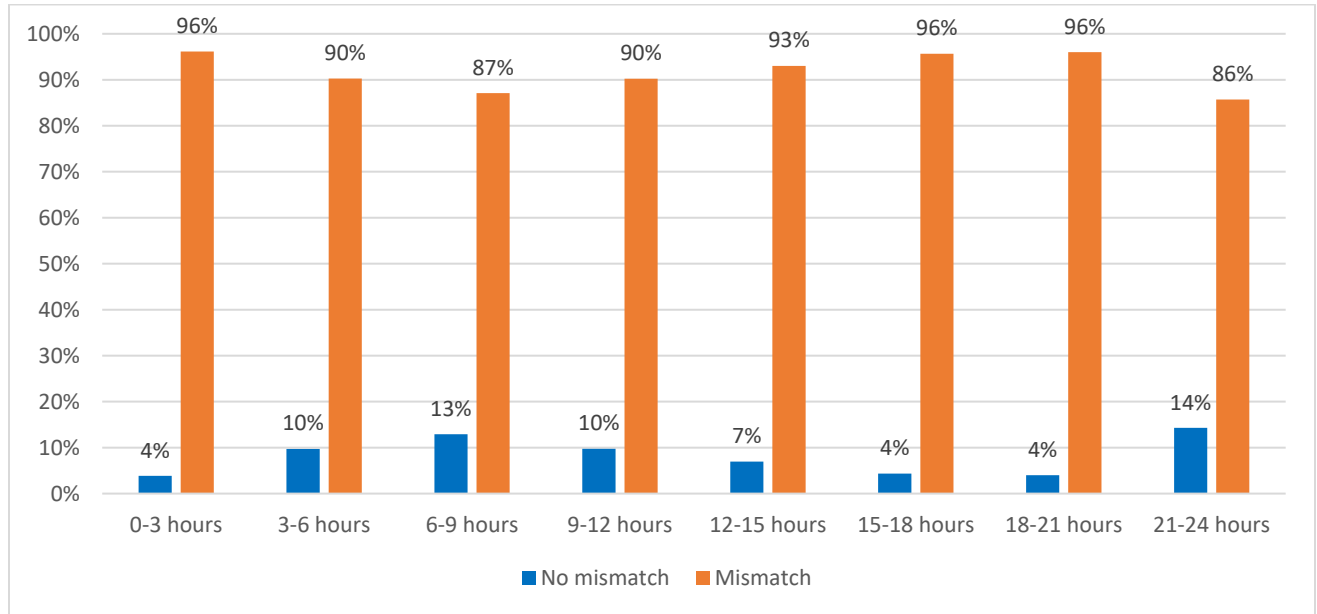
eFigure 4. Comparison of patients with CT hypodensity larger than CTP/MRI core volume(blue) and CTP/MRI core volumes larger than the CT hypodensity(orange) across time from LKW to randomization.

As time to randomization increased, proportion of patients demonstrating CT hypodensity larger than CTP/MRI core volume also increased – from 19% within 3 hours of LKW to 86% in those randomized 21-24 hours from LKW.

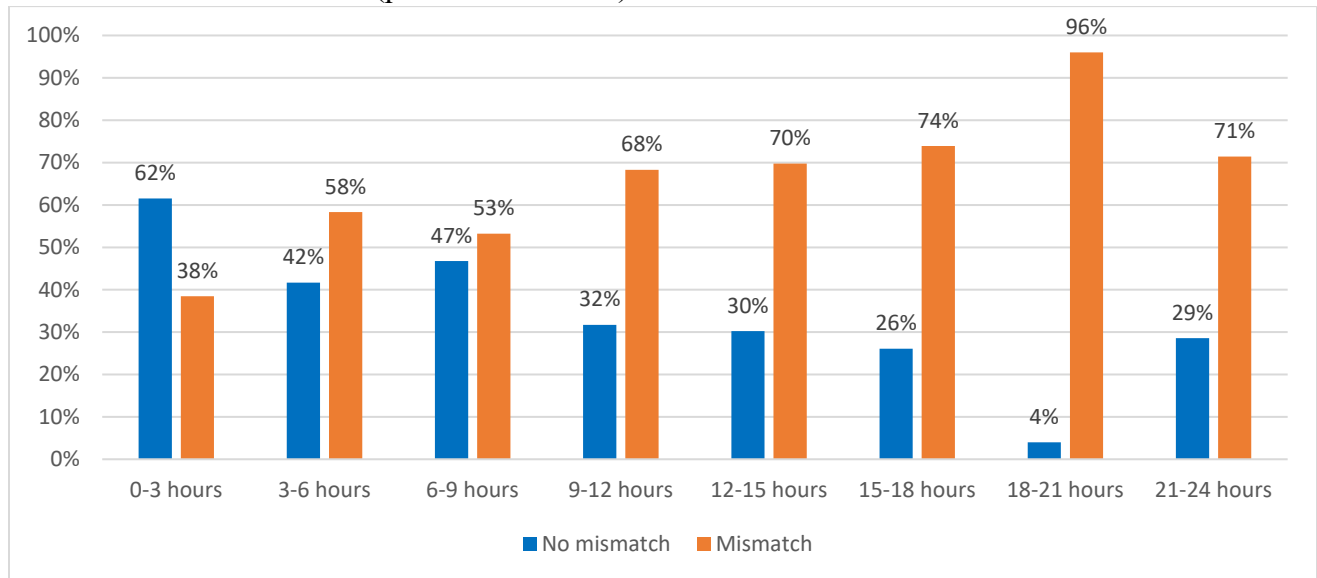


eFigure 5. Presence of penumbral patterns with time

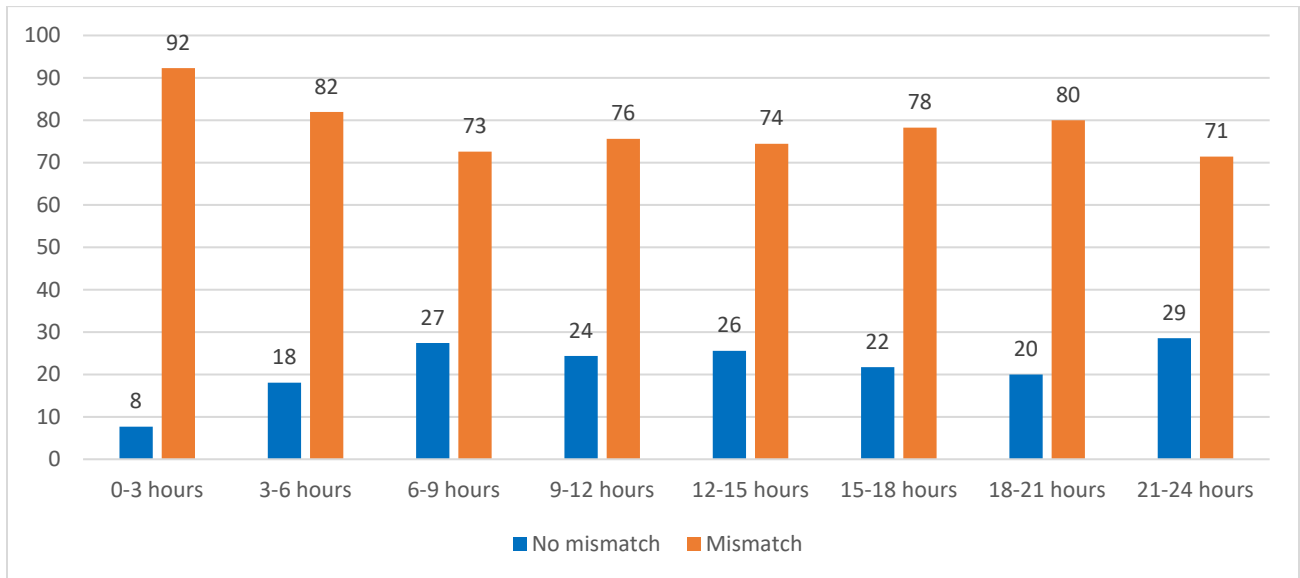
- a) Based on mismatch profile defined as a mismatch ratio ≥ 1.2 & mismatch volume $\geq 10\text{ml}$ from CTP/MRI core (p for trend=0.71)



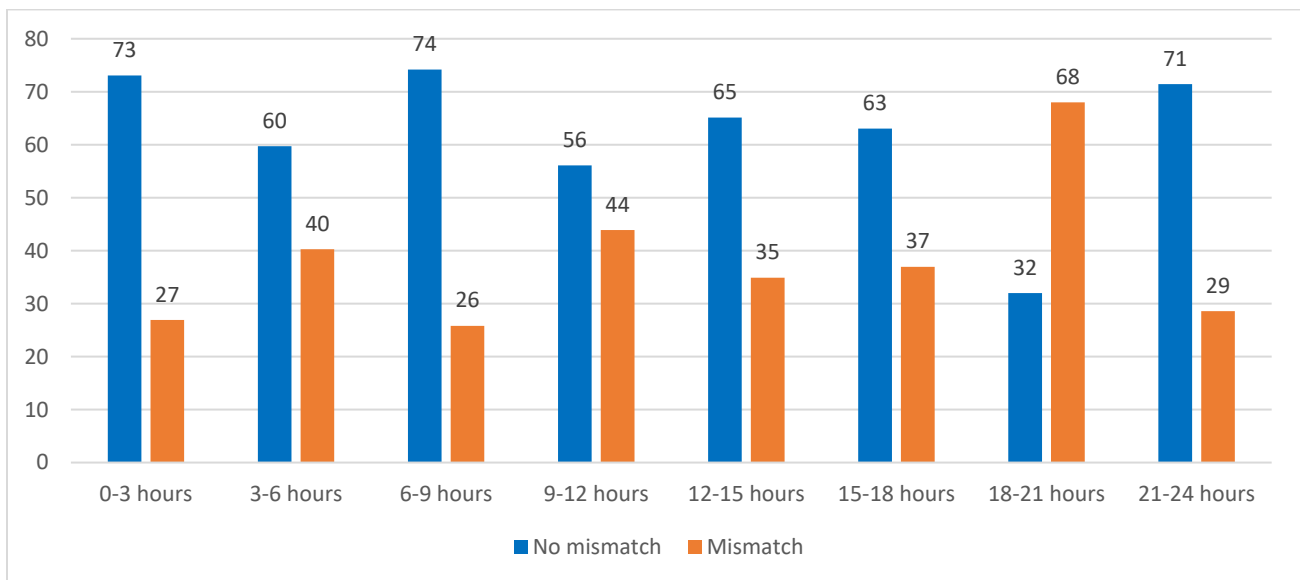
- b) Based on mismatch profile defined as a mismatch ratio ≥ 1.8 & mismatch volume $\geq 15\text{ml}$ from CTP/MRI core (p for trend <0.001)



- c) Based on mismatch profile defined as a mismatch ratio ≥ 1.2 & mismatch volume $\geq 10\text{ml}$ from Composite core (p for trend=0.27)

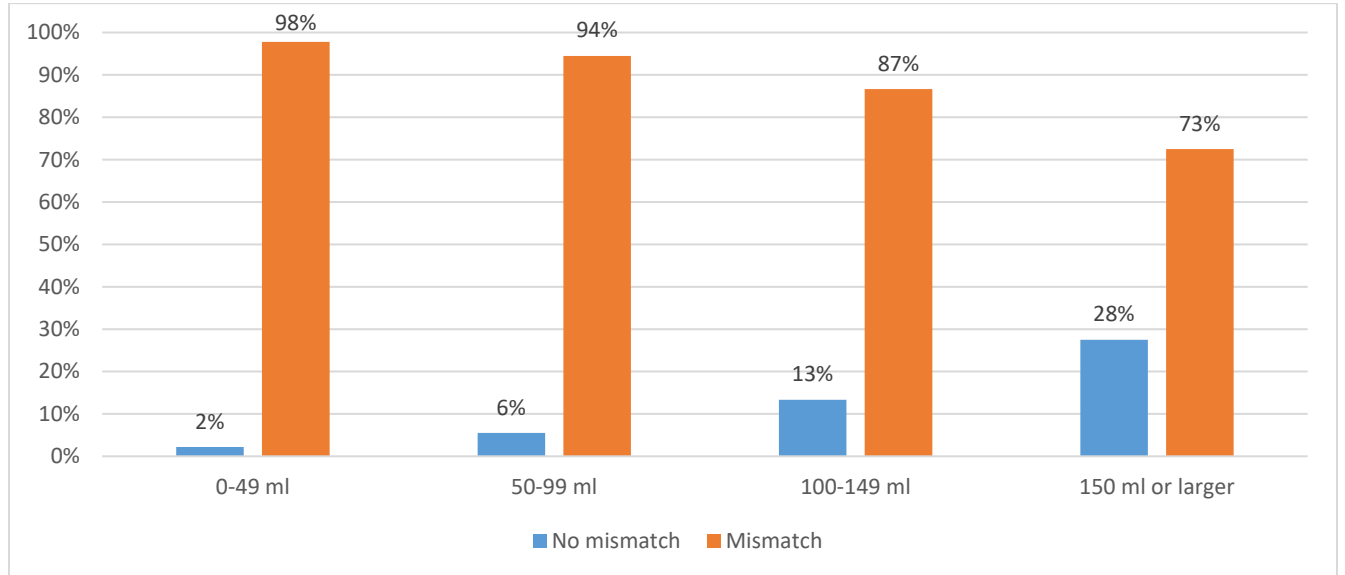


- d) Based on mismatch profile defined as a mismatch ratio ≥ 1.8 & mismatch volume $\geq 15\text{ml}$ from Composite core (p for trend=0.18)

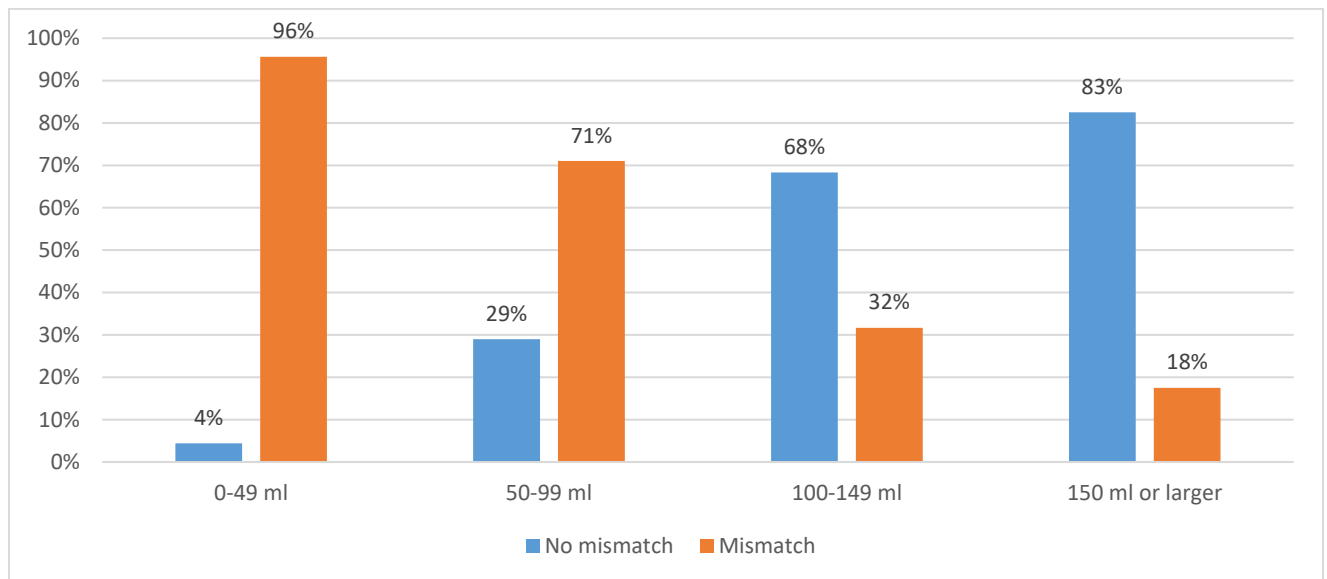


eFigure 6. Penumbral patterns by baseline CTP/MRI core estimates.

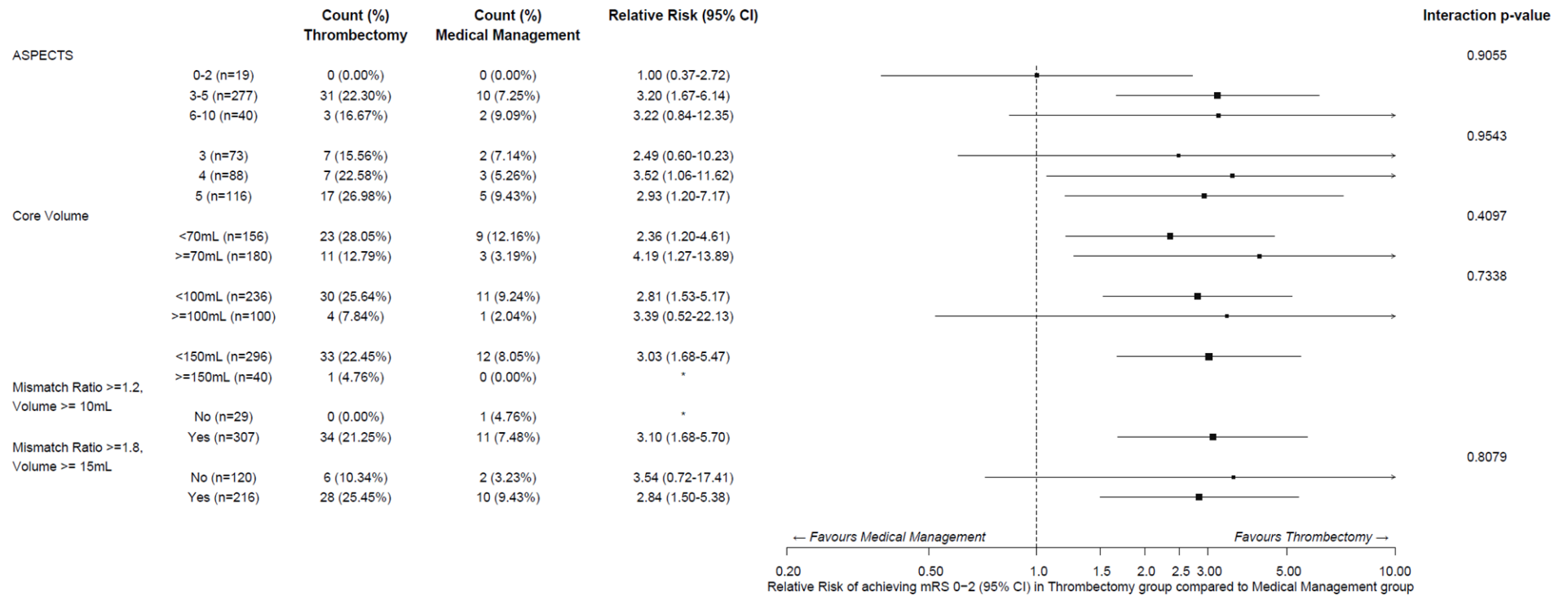
- a) Based on mismatch profile defined as a mismatch ratio ≥ 1.2 & mismatch volume ≥ 10 ml from CTP/MRI core. As baseline core volume increased, proportion of patients with no mismatch increased (p for trend < 0.001)



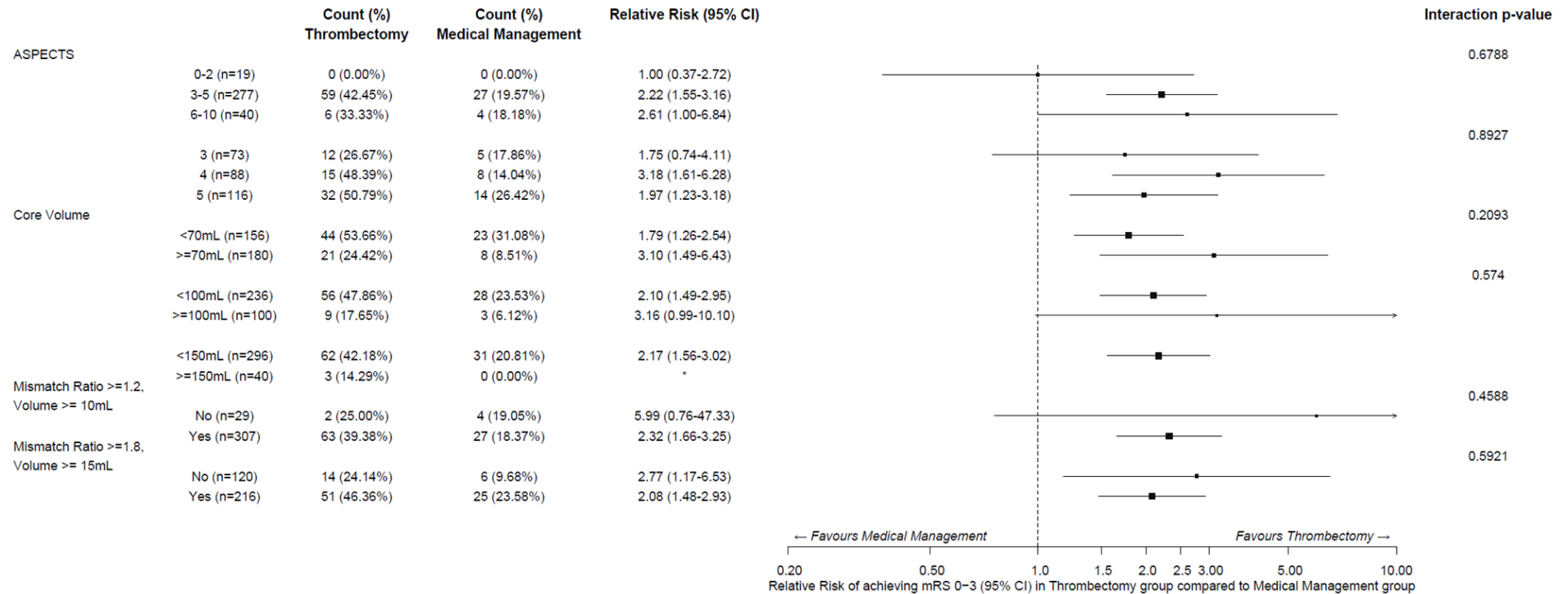
- b) Based on mismatch profile defined as a mismatch ratio ≥ 1.8 & mismatch volume ≥ 15 ml from CTP/MRI core. As baseline core volume increased, proportion of patients with no mismatch increased (p for trend < 0.001)



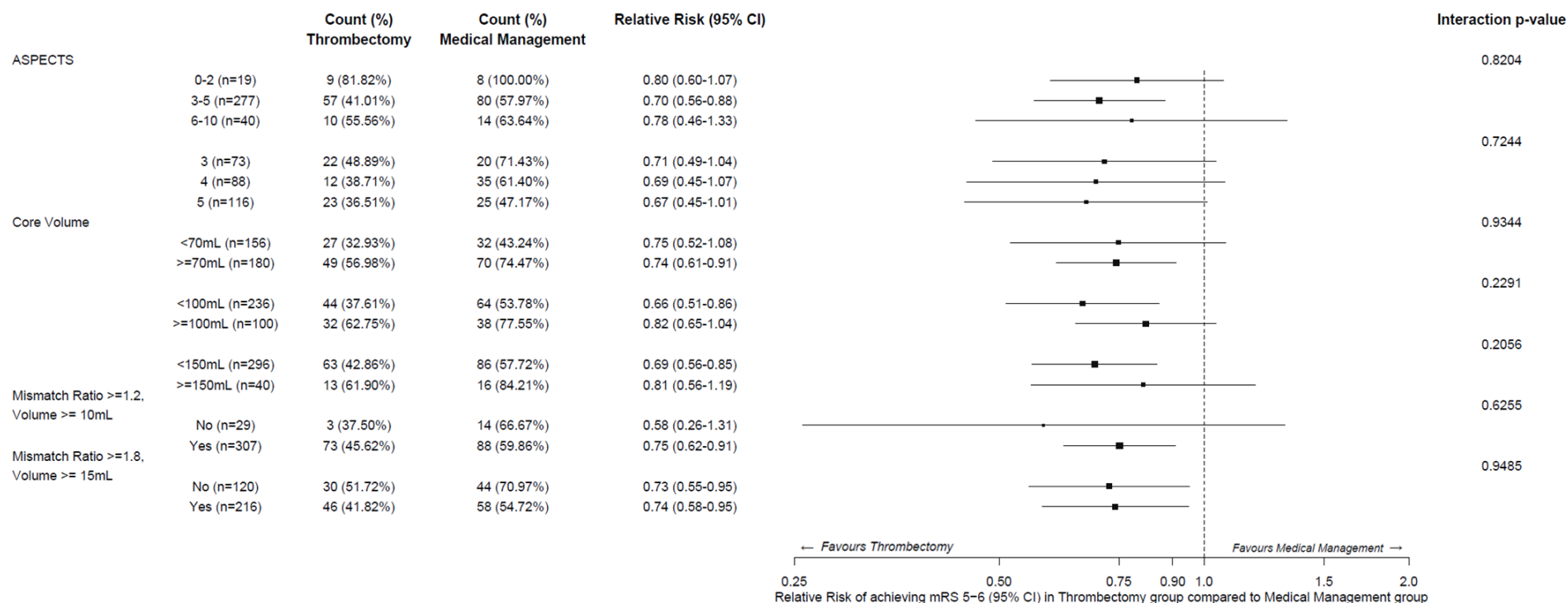
eFigure 7. Forest plot illustration for EVT treatment effect on functional independence(mRS 0-2) at 90-day follow-up across various imaging strata.



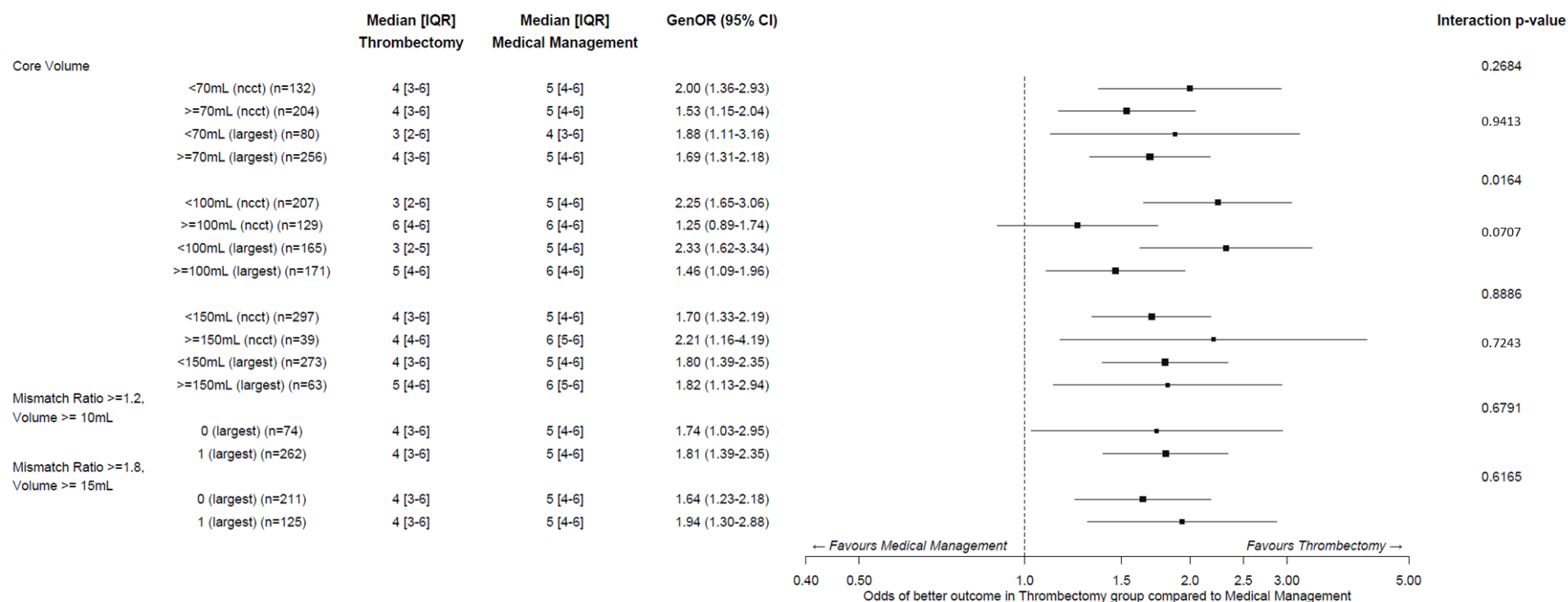
eFigure 8. Forest plot illustration for EVT treatment effect on independent ambulation(mRS 0-3) at 90-day follow-up across various imaging strata.



eFigure 9. Forest plot illustration for EVT treatment effect on complete dependence or death(mRS 5-6) at 90-day follow-up across various imaging strata.



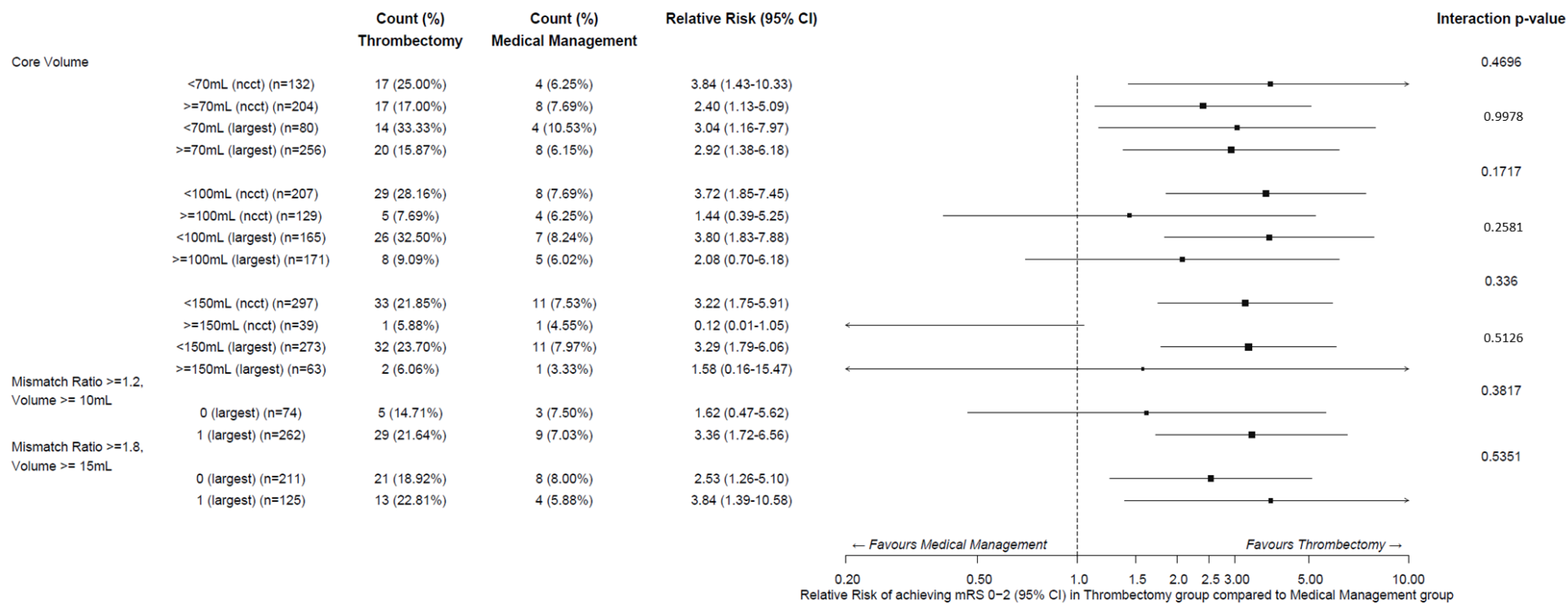
eFigure 10. Forest plot illustration for EVT treatment effect on functional outcome (modified Rankin Scale Score) at 90-day follow-up across various imaging strata.



Largest represents composite core volume.

NCCT represents CT hypodensity volume

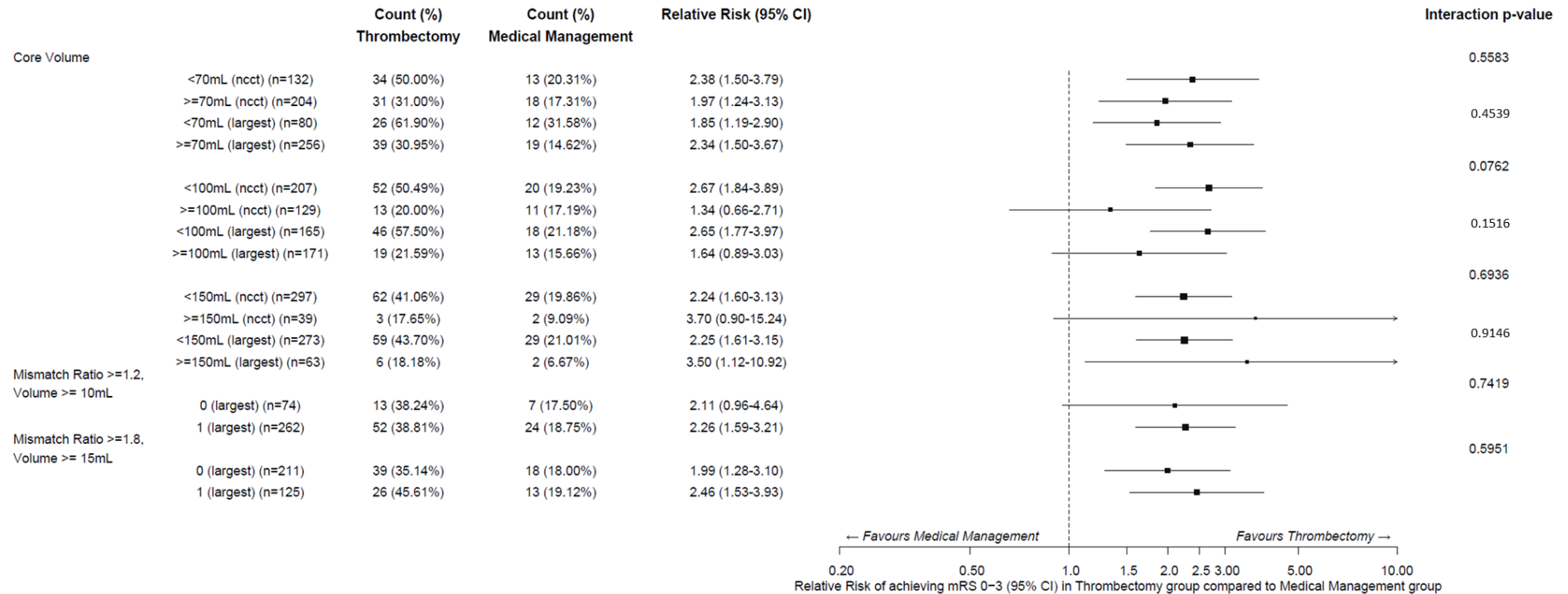
eFigure 11. Forest plot illustration for EVT treatment effect on functional independence(mRS 0-2) at 90-day follow-up across various imaging strata.



Largest represents composite core volume.

NCCT represents CT hypodensity volume

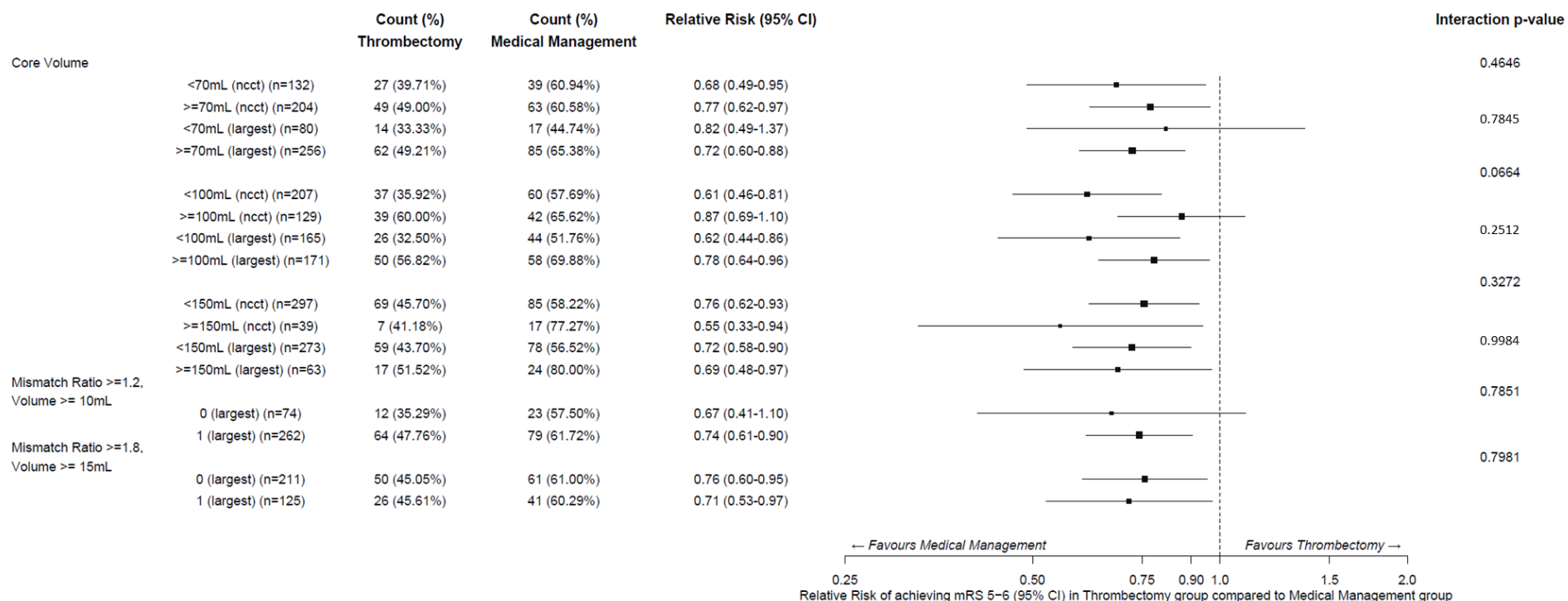
eFigure 12. Forest plot illustration for EVT treatment effect on independent ambulation(mRS 0-3) at 90-day follow-up across various imaging strata.



Largest represents composite core volume.

NCCT represents CT hypodensity volume

eFigure 13. Forest plot illustration for EVT treatment effect on complete dependence or death(mRS 5-6) at 90-day follow-up across various imaging strata.

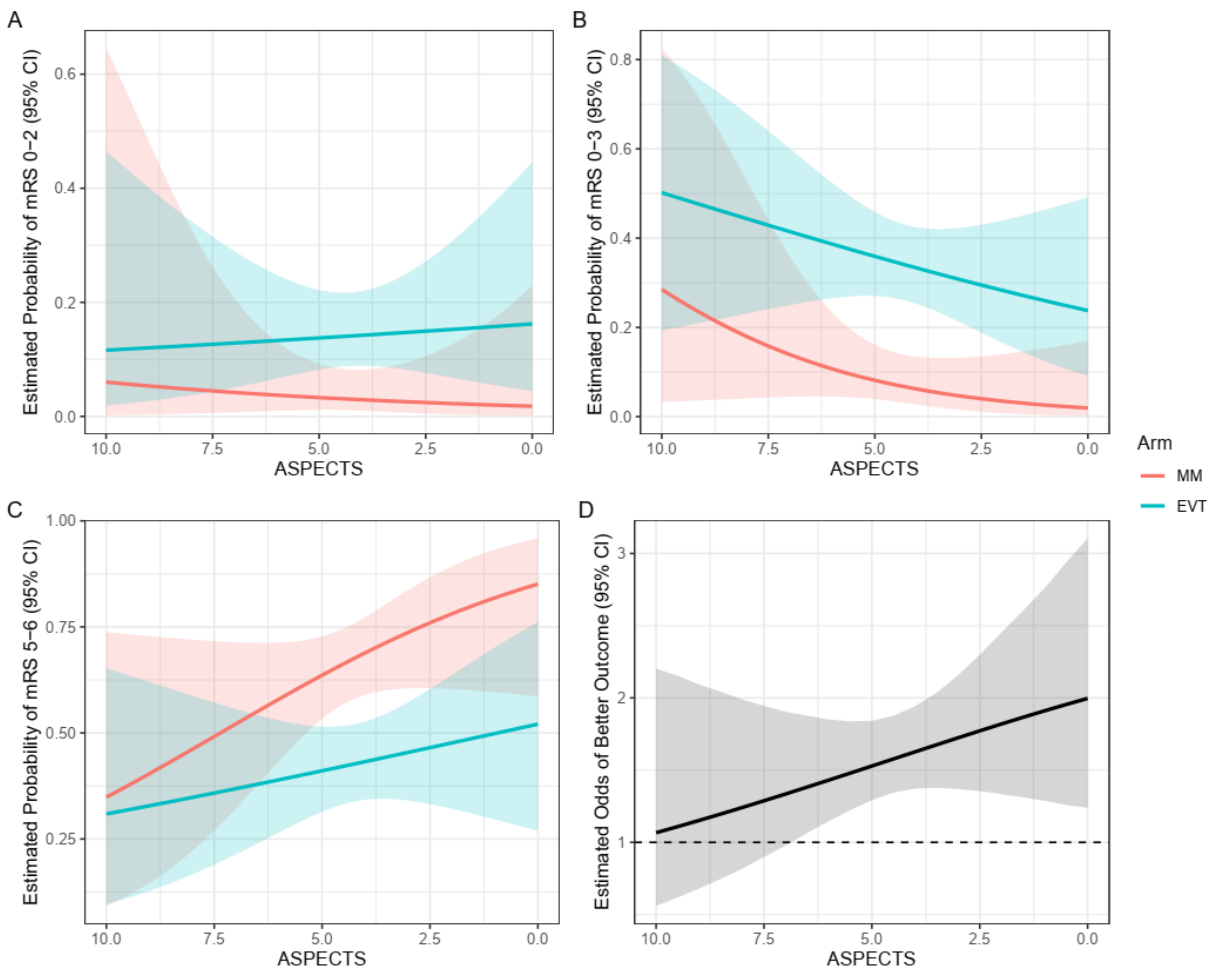


Largest represents composite core volume.

NCCT represents CT hypodensity volume

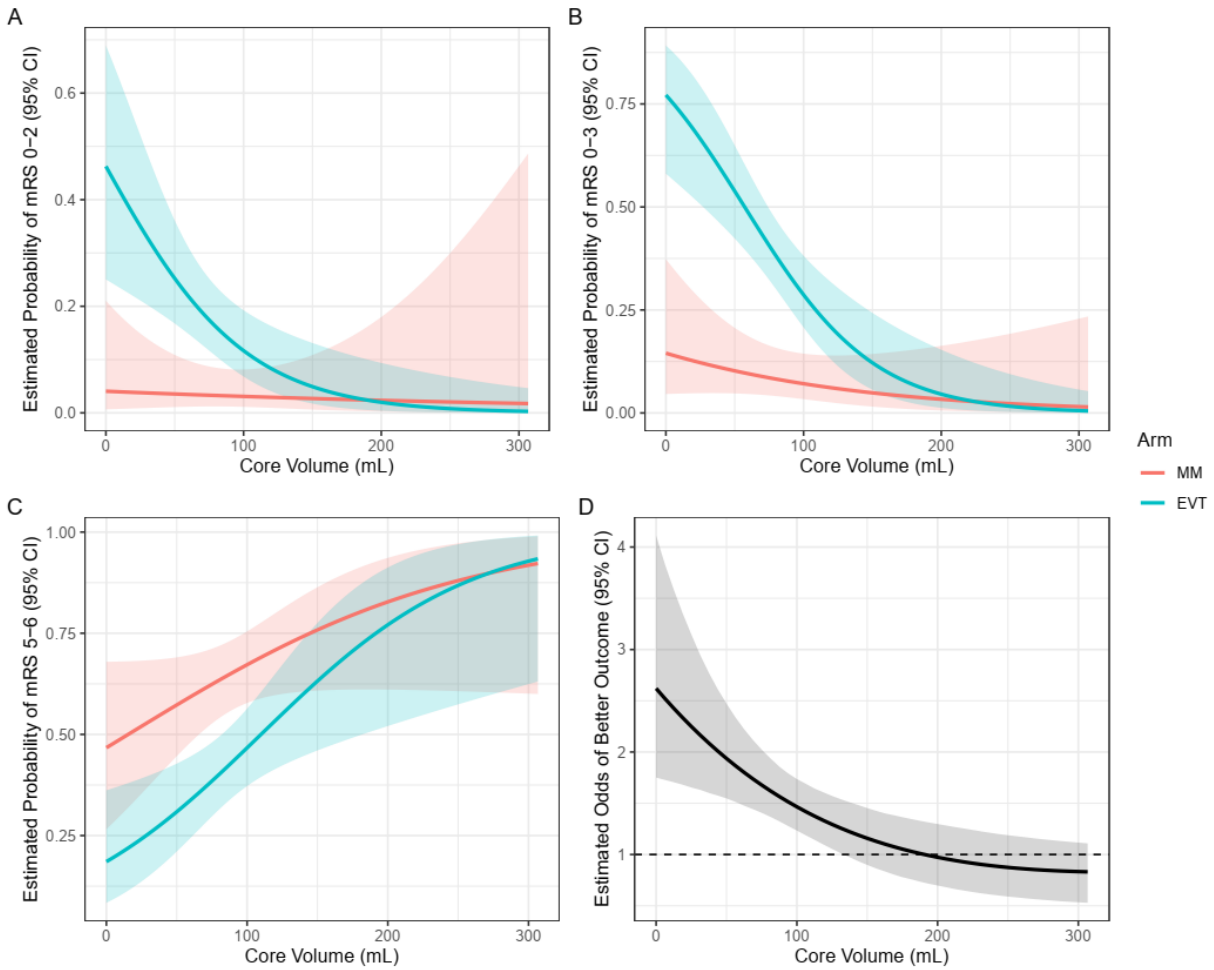
eFigure 14. Association of decreasing CT ASPECTS score and functional outcomes in patients receiving EVT and MM(adjusted for composite core volume).

Panels A, B, and C– illustrate estimated probability of mRS 0-2, 0-3 and 5-6, respectively. Panel D illustrates the Generalised Odds of having a ≥ 1 point improvement in mRS outcome as a function of CT ASPECTS. Please note that patients with ASPECTS 6-10 had to have large core on CT-perfusion to be eligible for SELECT2, which may explain the small treatment effect in patients with favorable ASPECTS compared to patients with lower ASPECTS.



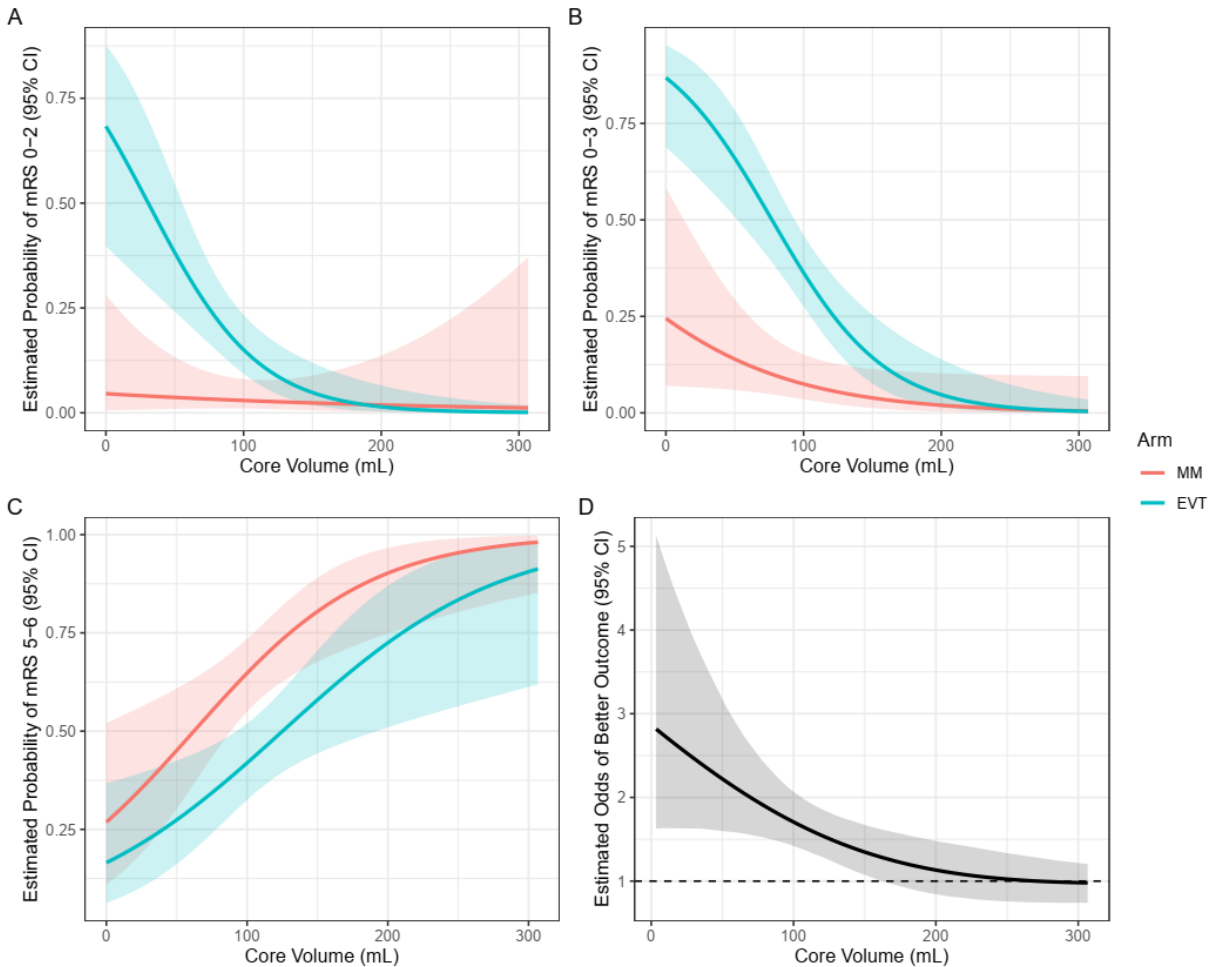
eFigure 15. Association of increasing CT hypodensity volume and functional outcomes in patients receiving EVT and MM.

Panel A, B, and C—illustrate estimated probability of mRS 0-2, 0-3 and 5-6, respectively. Panel D illustrates the Generalised Odds of having a ≥ 1 point improvement in mRS outcome as a function of CT hypodensity volume.



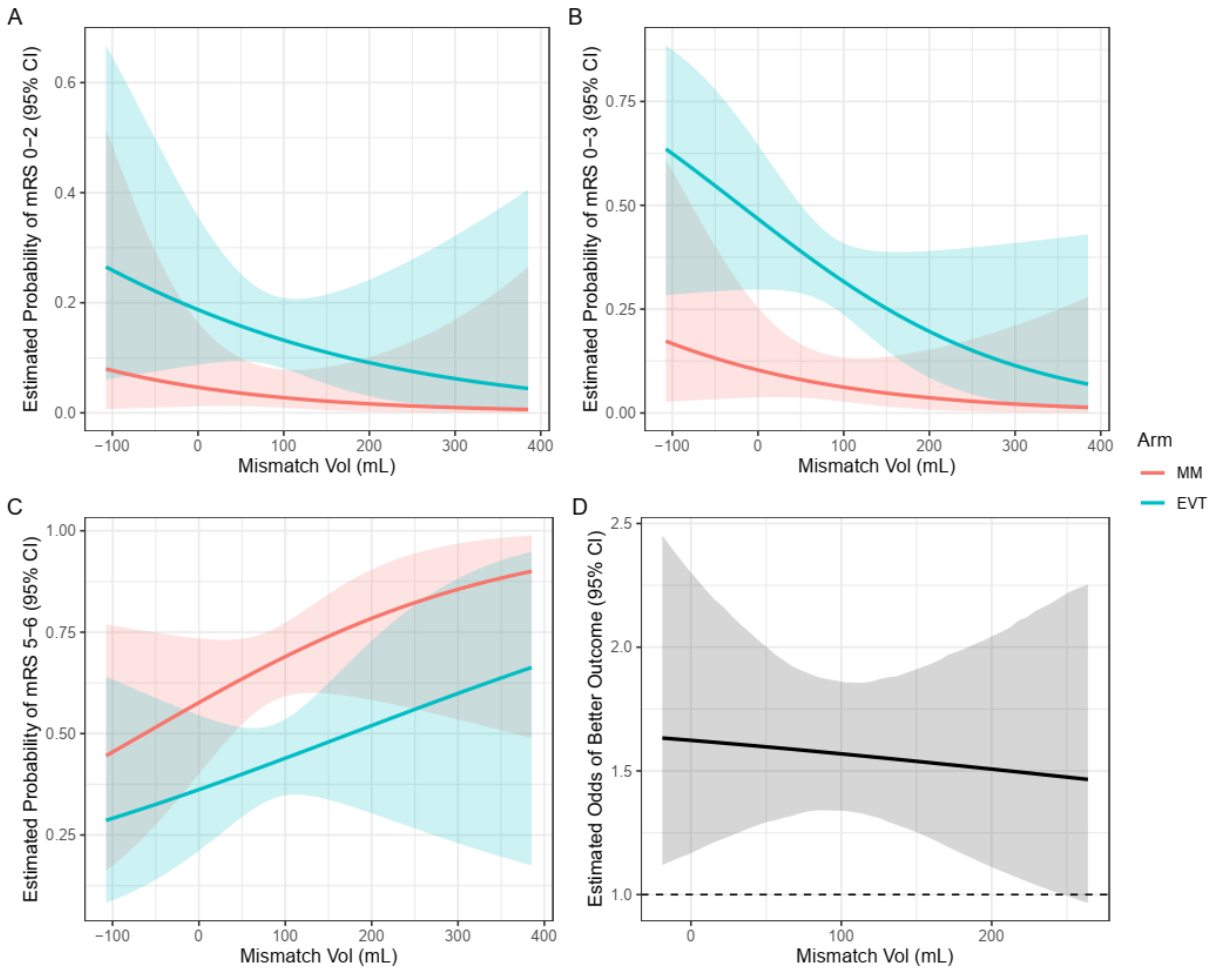
eFigure 16. Association of increasing composite core volume and functional outcomes in patients receiving EVT and MM.

Panels A, B, and C— illustrate estimated probability of mRS 0-2, 0-3 and 5-6, respectively. Panel D illustrates the Generalised Odds of having a ≥ 1 point improvement in mRS outcome as a function of composite core volume.



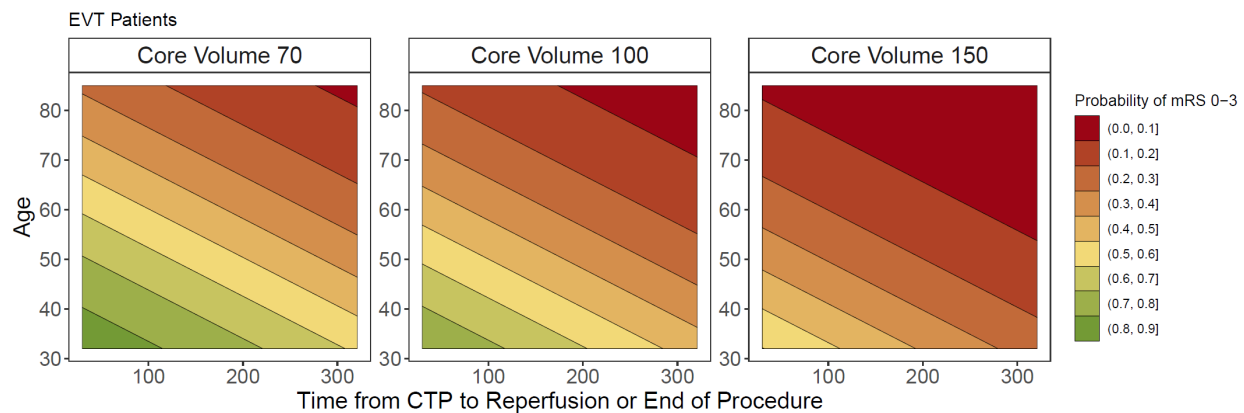
eFigure 17. Association of increasing mismatch volume and functional outcomes in patients receiving EVT and MM.

Panels A, B, and C– illustrate estimated probability of mRS 0-2, 0-3 and 5-6, respectively. Panel D illustrates the Generalised Odds of having a ≥ 1 point improvement in mRS outcome as a function of mismatch volume.



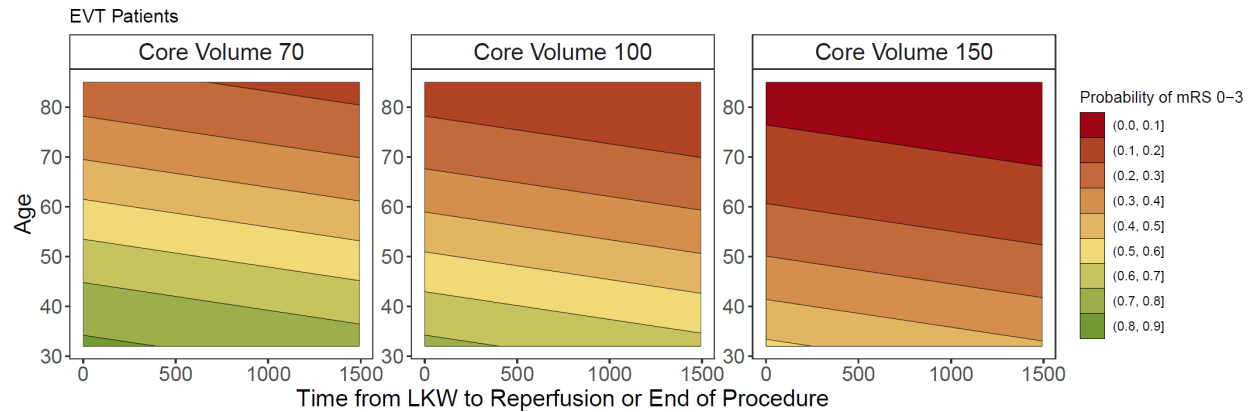
eFigure 18 Illustrative predicted probabilities of independent ambulation(mRS 0-3) and how they change with increasing age and time from perfusion imaging acquisition to reperfusion/end of the procedure with CTP/MRI core volume set at a) 70ml, b) 100ml and c) 150ml.

As age and time from perfusion imaging acquisition increases, the predicted probability of independent ambulation decreases. This relationship is consistent across composite core volume estimates, but overall predicted probability decreases as composite core volume estimate increases. As age increases and time from imaging to reperfusion/end of the procedure lengthens, the probability of independent ambulation decreases, such that it is very unlikely that a patient with 150mL core would achieve independent ambulation unless they are young and treated very rapidly.



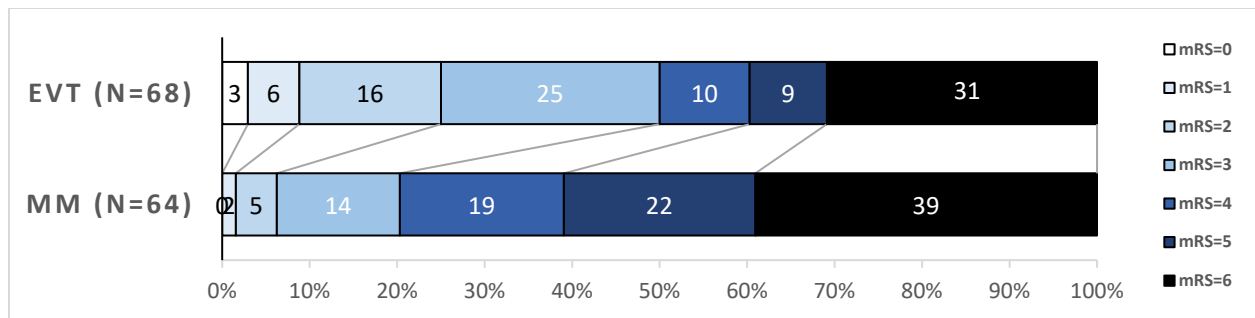
eFigure 19. Illustrative predicted probabilities of independent ambulation(mRS 0-3) and how they change with increasing age and time from last known well to reperfusion/end of the procedure with composite core volume set at a) 70ml, b) 100ml and c) 150ml.

As age and time from LKW to reperfusion/end of the procedure increases, the predicted probability of independent ambulation decreases. This relationship is consistent across composite core volume estimates, but overall predicted probability decreases as composite core volume estimate increases.

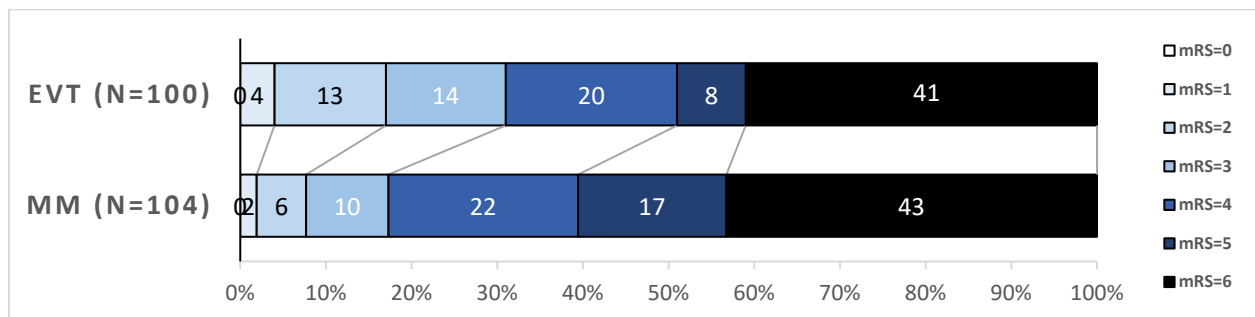


eFigure 20. Distribution of modified Rankin Scale scores by treatment across different imaging strata.

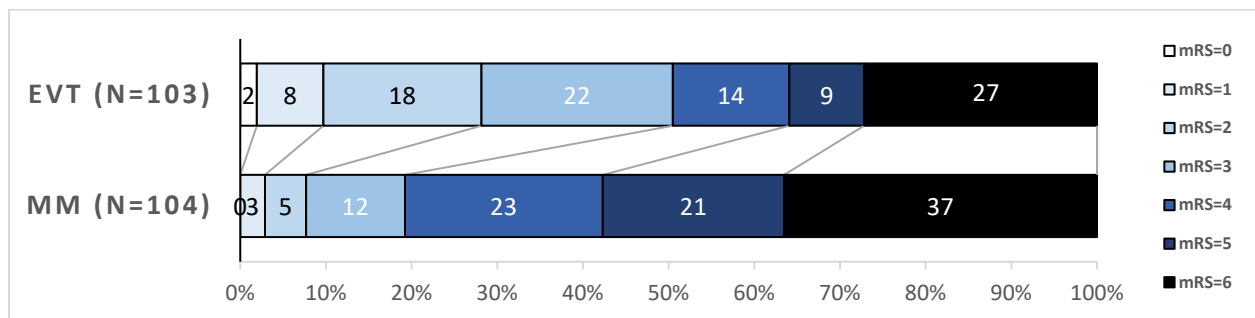
a) In patients with CT hypodensity volume <70ml



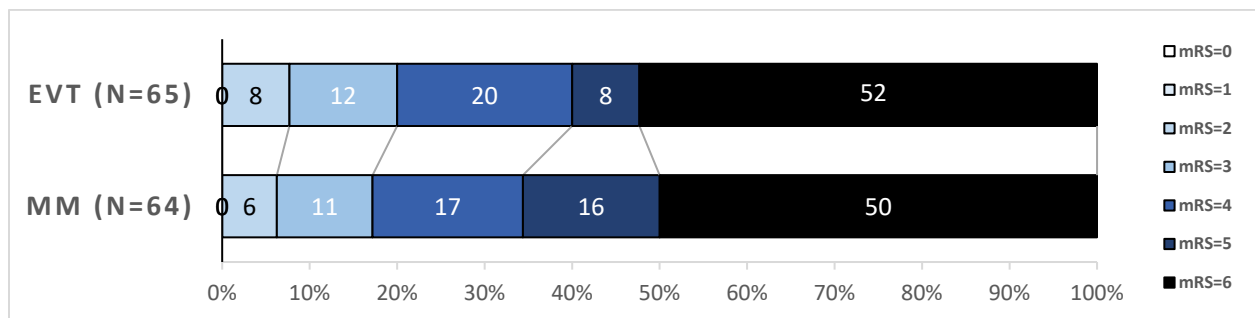
b) in patients with CT hypodensity ≥70ml



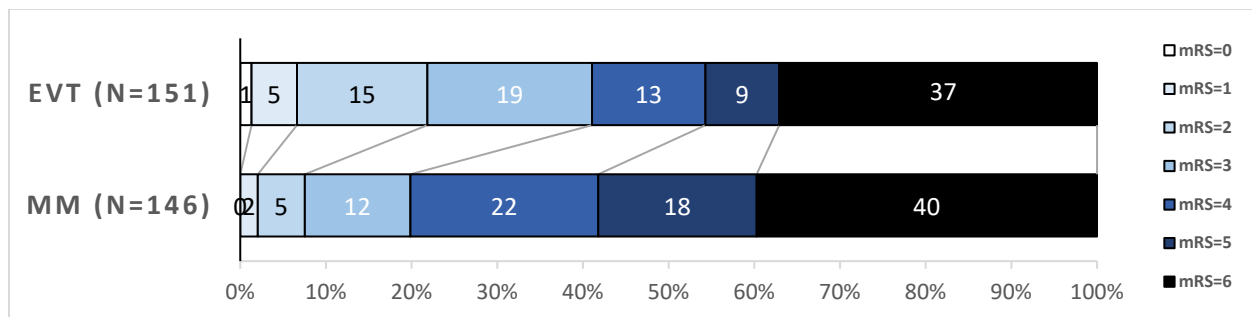
c) in patients with CT hypodensity <100ml



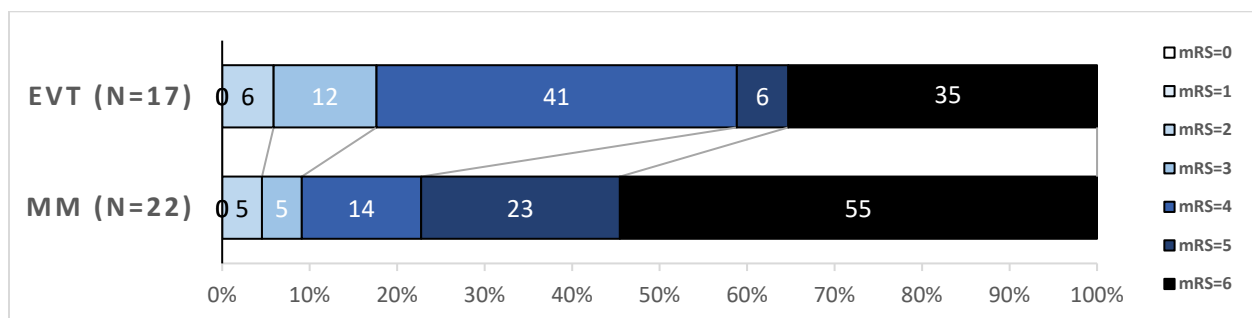
d) in patients with CT hypodensity ≥100ml



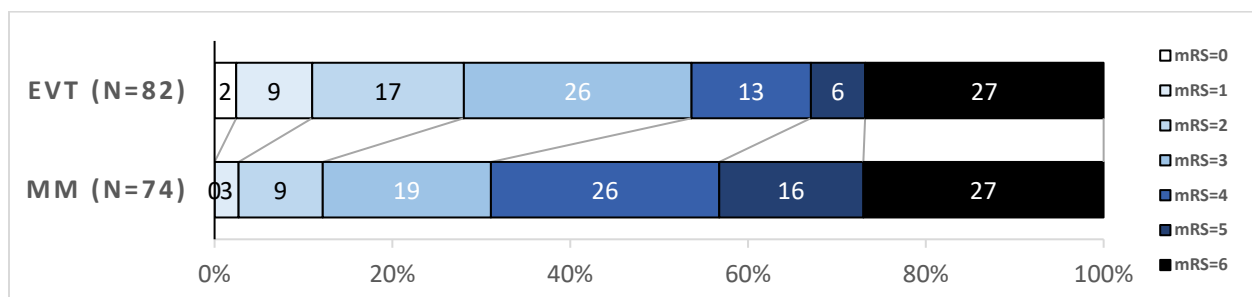
e) in patients with CT hypodensity <150ml



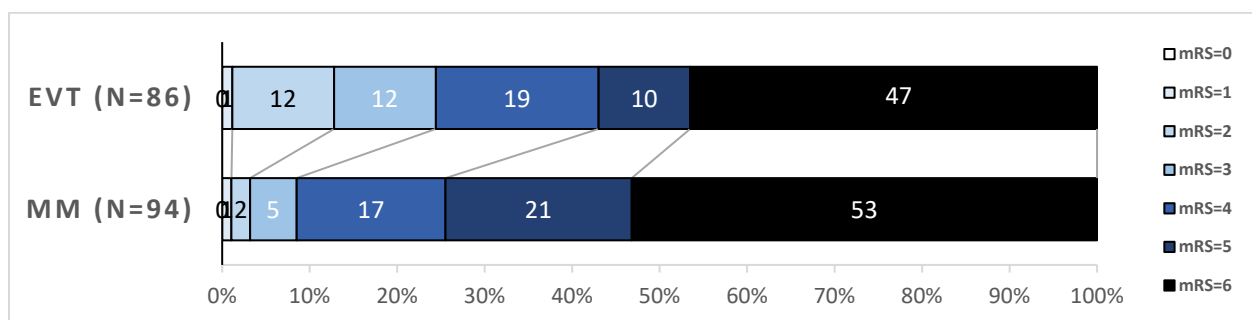
f) in patients with CT hypodensity $\geq 150\text{ml}$



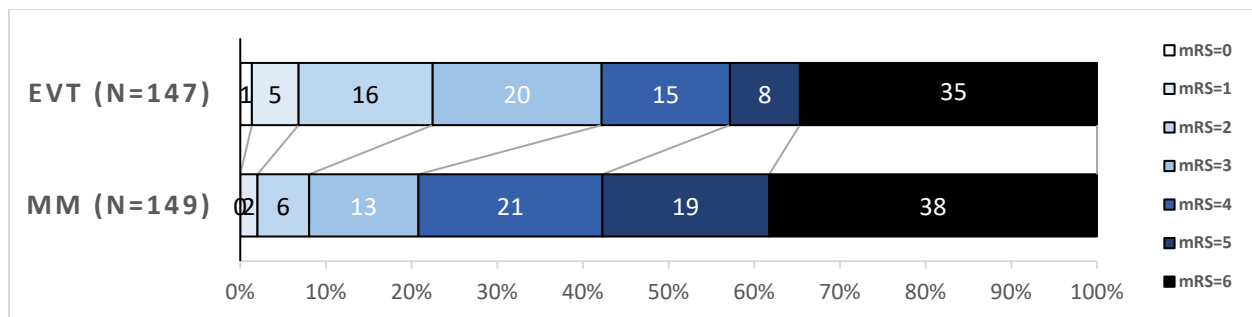
g) in patients with CTP/MRI core $< 70\text{ml}$



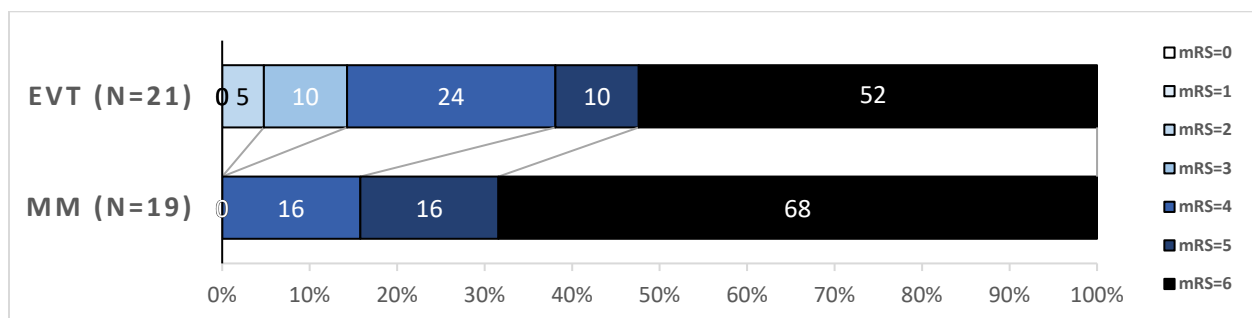
h) in patients with CTP/MRI core $\geq 70\text{ml}$



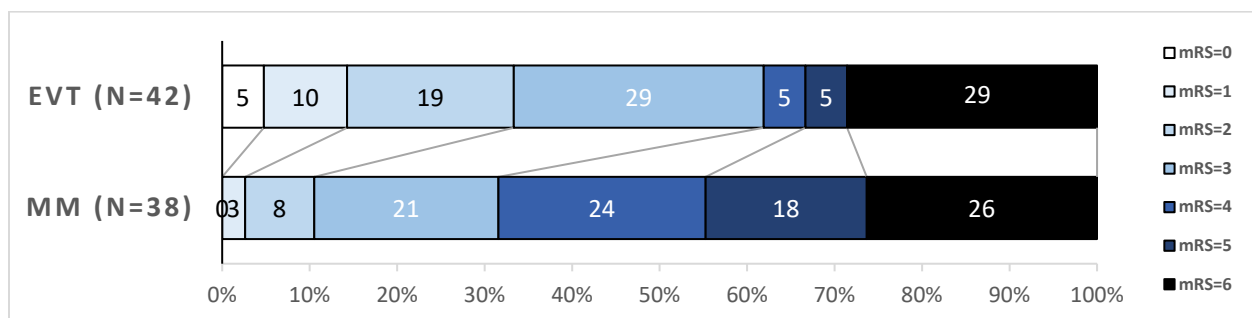
i) in patients with CTP/MRI $< 150\text{ml}$



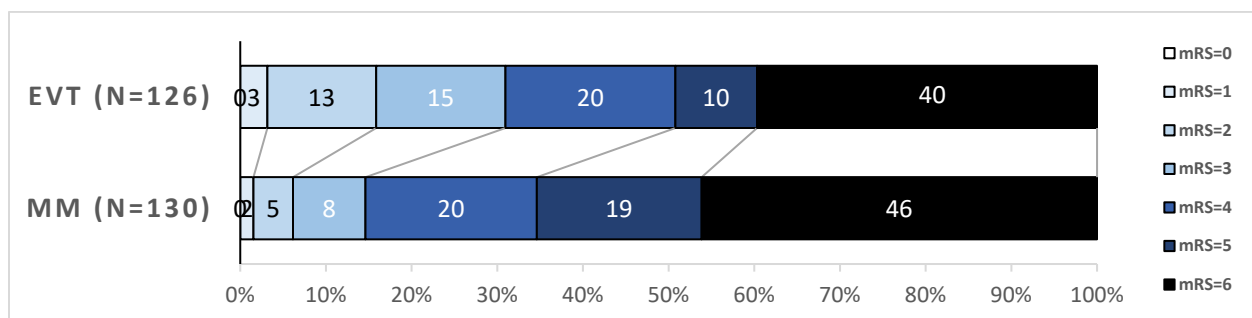
j) in patients with CTP/MRI core $\geq 150\text{ml}$



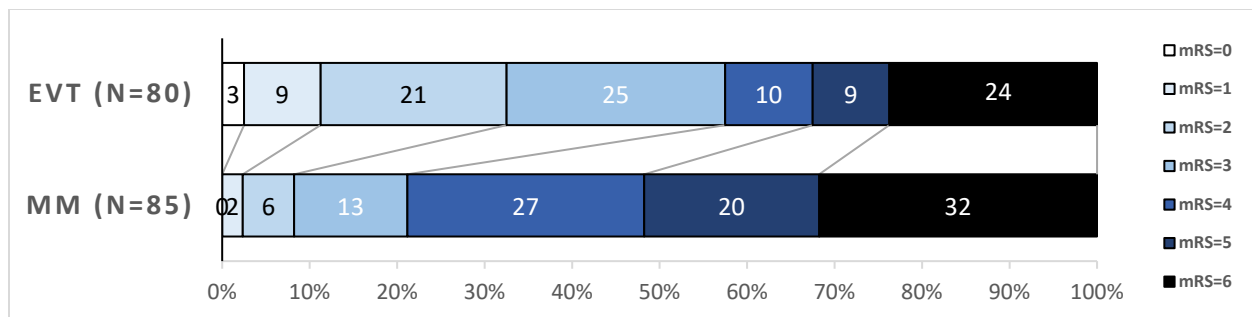
k) in patients with composite core $< 70\text{ml}$



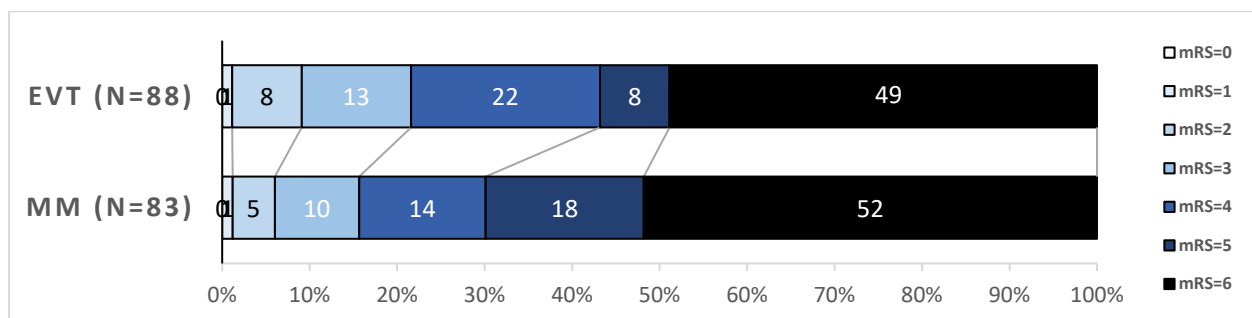
l) in patients with composite core $\geq 70\text{ml}$



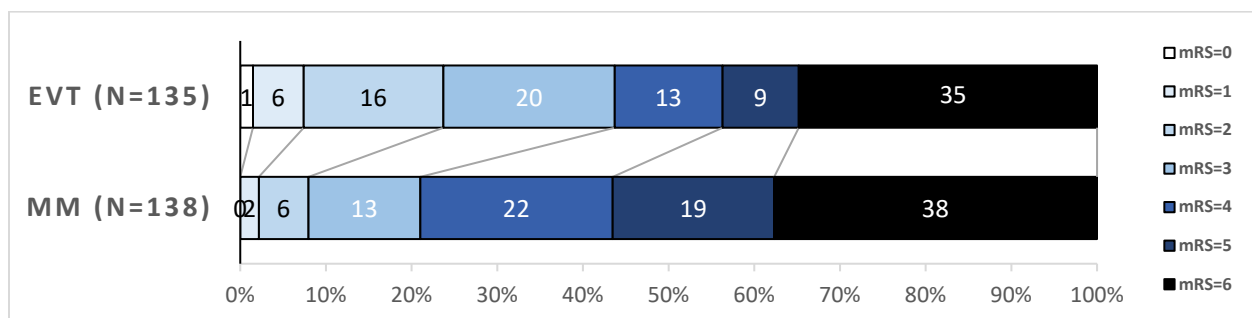
m) in patients with composite core $< 100\text{ml}$



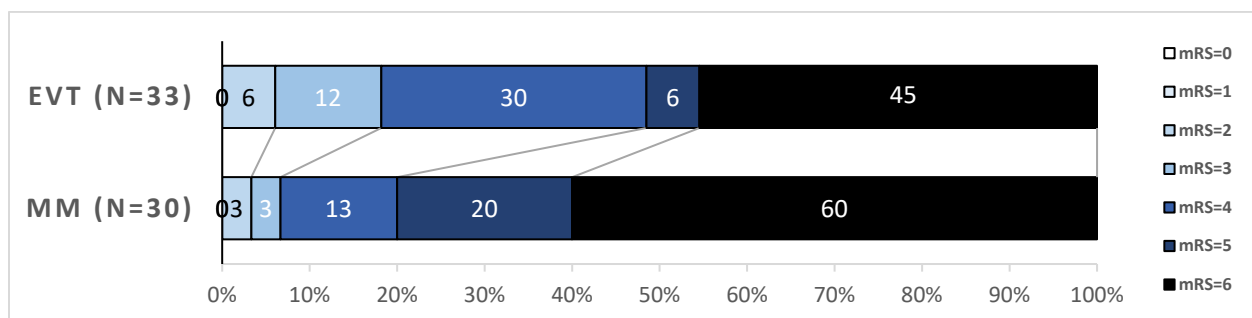
n) in patients with composite core $\geq 100\text{ml}$



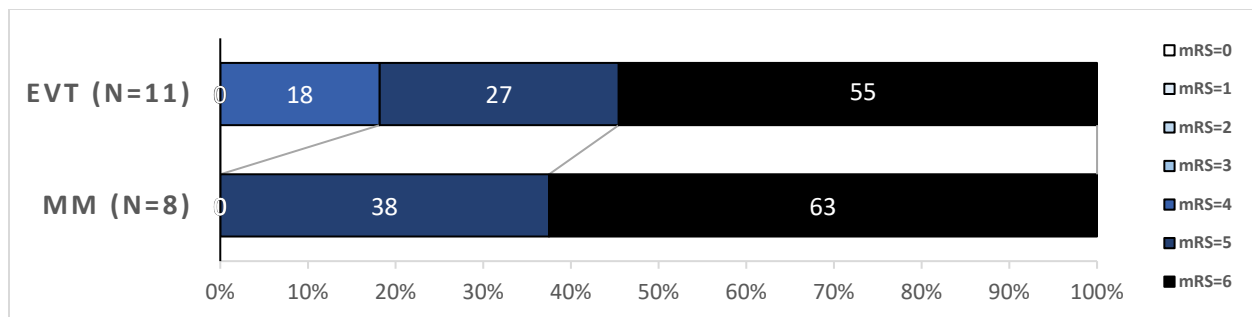
o) in patients with composite core $< 150\text{ml}$



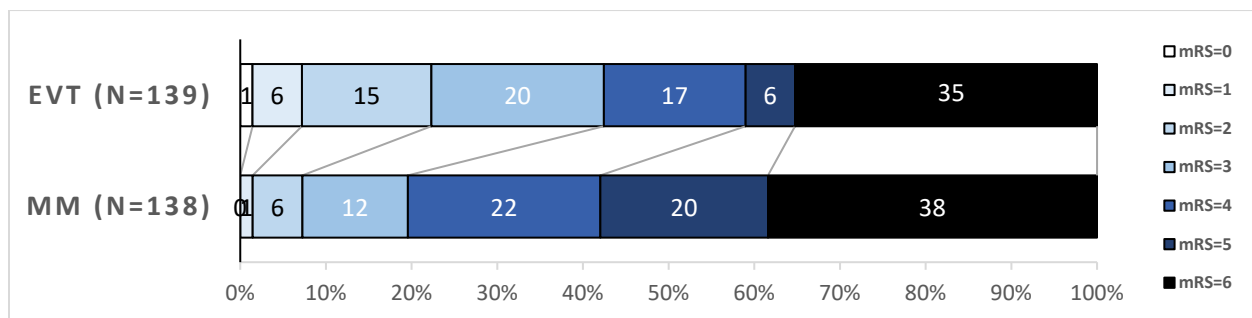
p) in patients with composite core $\geq 150\text{ml}$



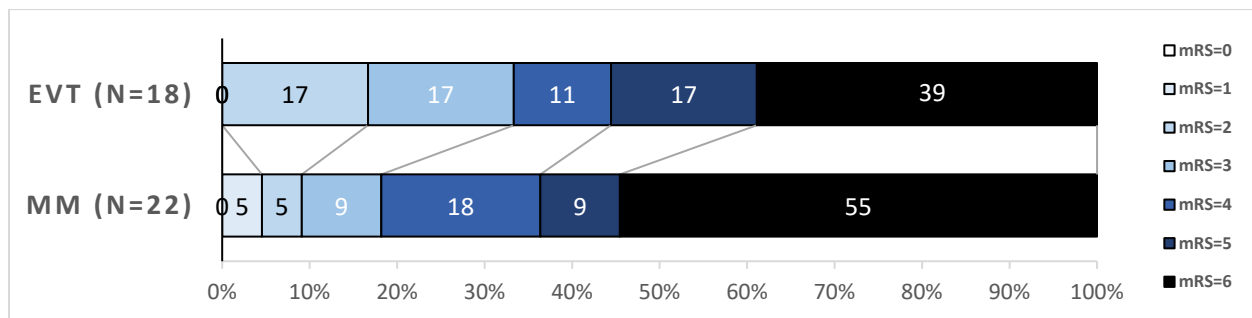
q) in patients with ASPECTS 0-2:



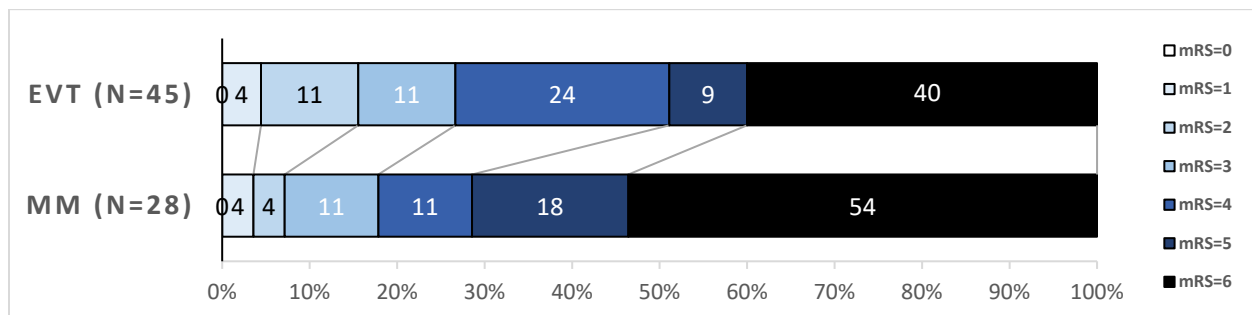
r) in patients with ASPECTS 3-5:



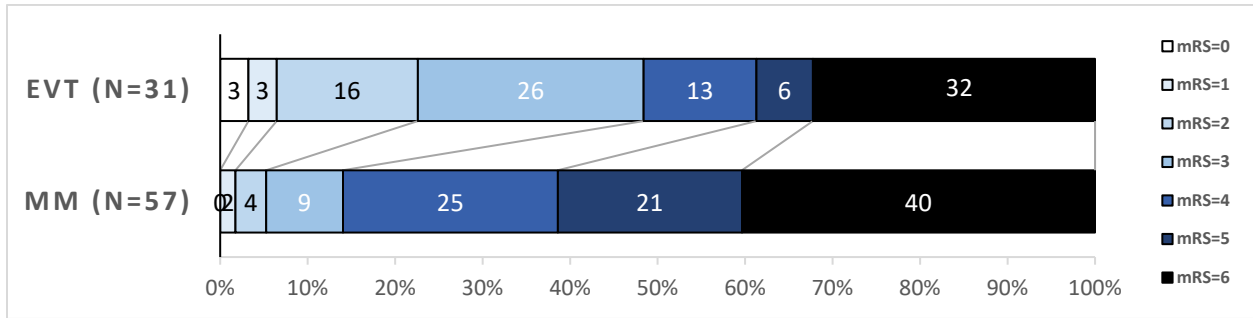
s) in patients with ASPECTS 6-10:



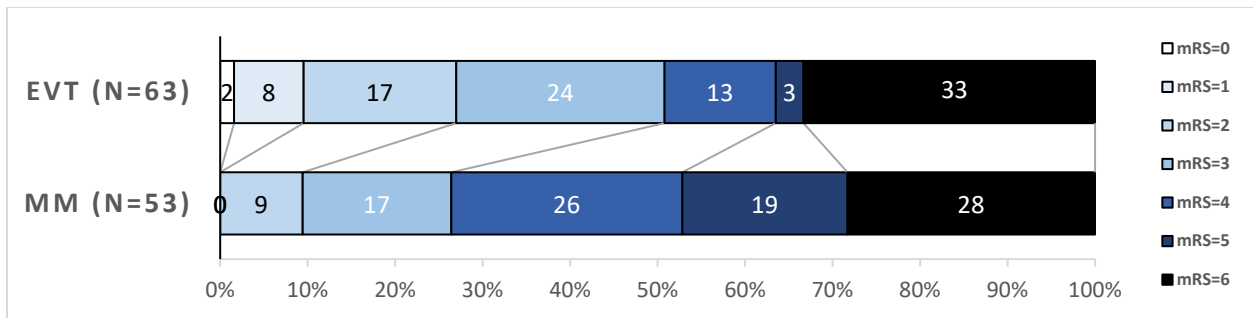
t) in patients with ASPECTS 3:



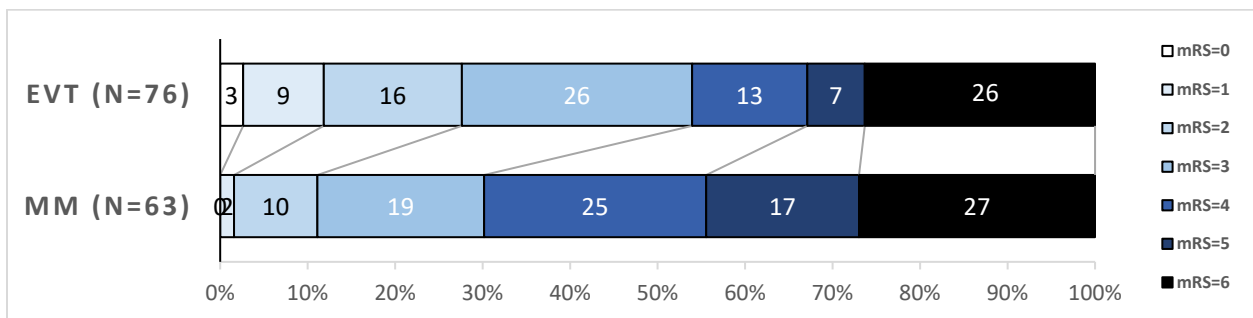
u) in patients with ASPECTS 4:



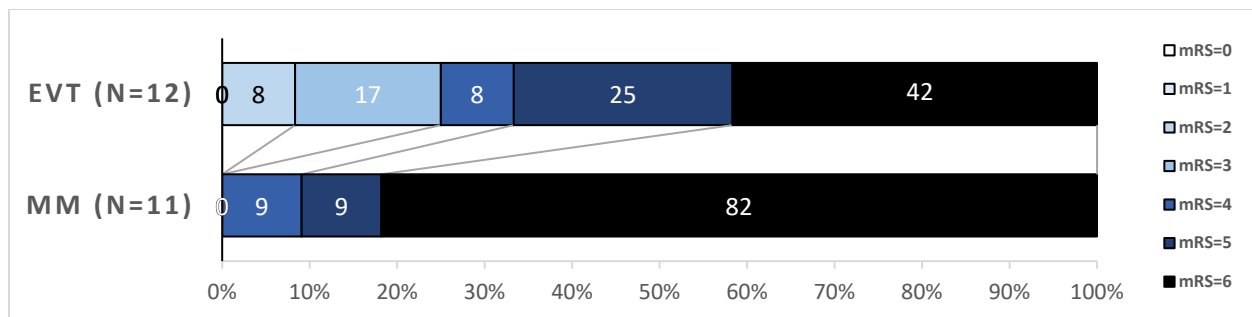
v) in patients with ASPECTS 5:



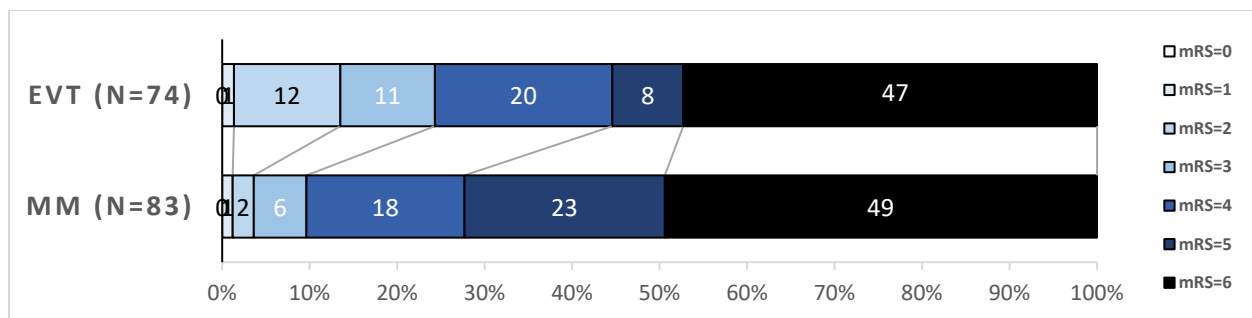
w) in patients with ASPECTS <6 & CTP/MRI core <70ml:



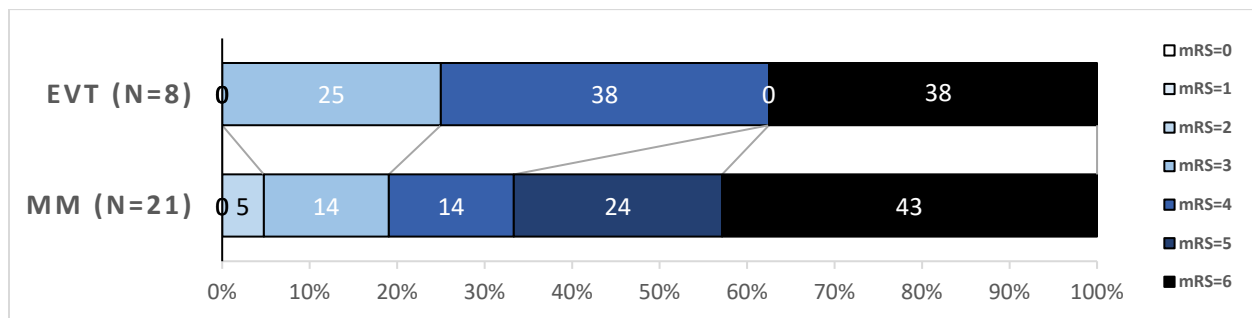
x) in patients with ASPECTS ≥6 & CTP/MRI core ≥70ml:



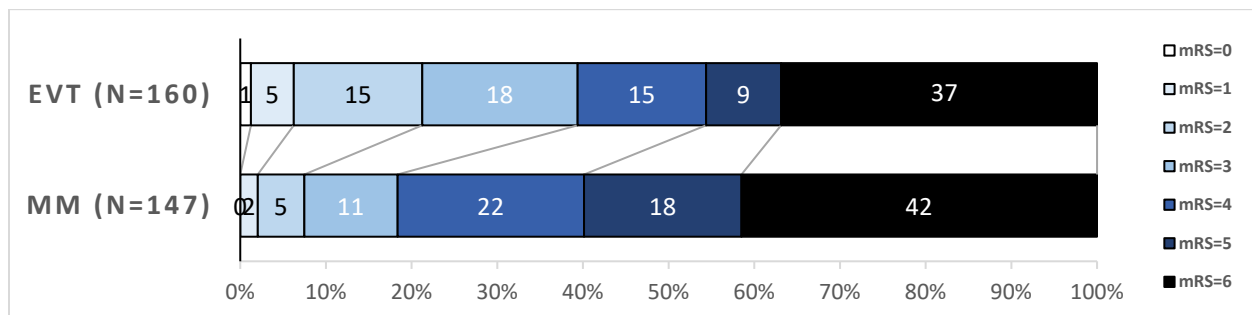
y) in patients with ASPECTS <6 & CTP/MRI core ≥ 70 ml:



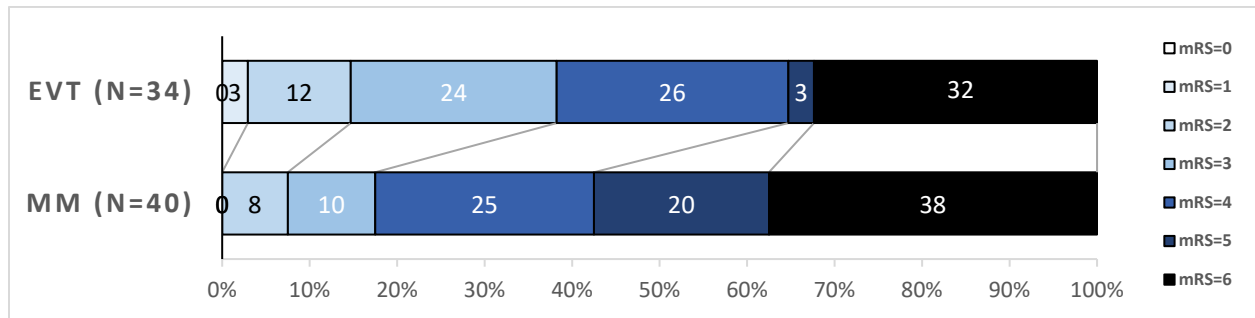
z) in patients without mismatch ratio ≥ 1.2 & mismatch volume ≥ 10 ml based on CTP/MRI core:



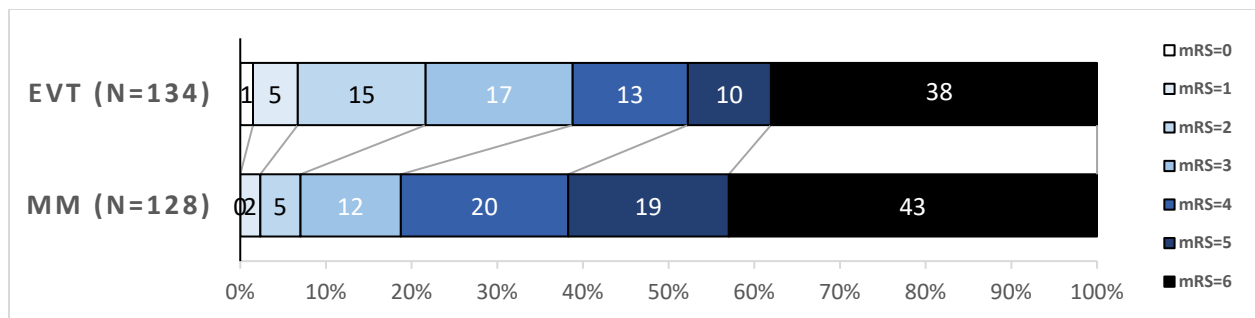
aa) in patients with mismatch ratio ≥ 1.2 & mismatch volume ≥ 10 ml based on CTP/MRI core:



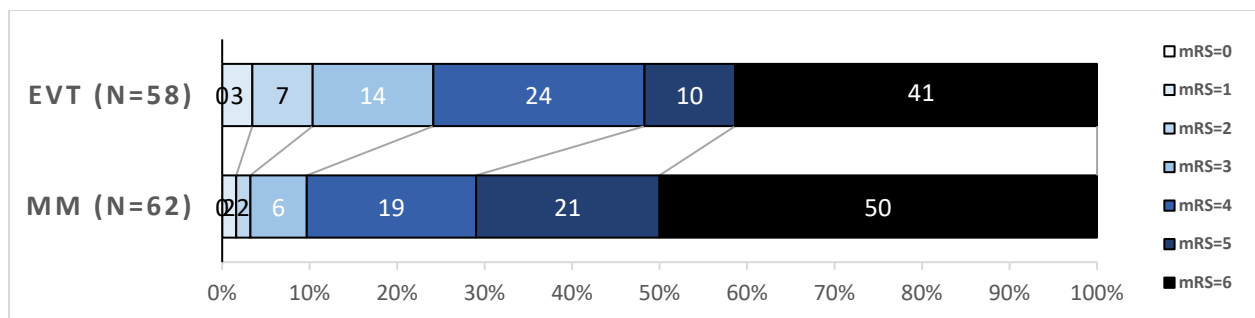
ab) in patients without mismatch ratio ≥ 1.2 & mismatch volume $\geq 10\text{ml}$ based on composite core:



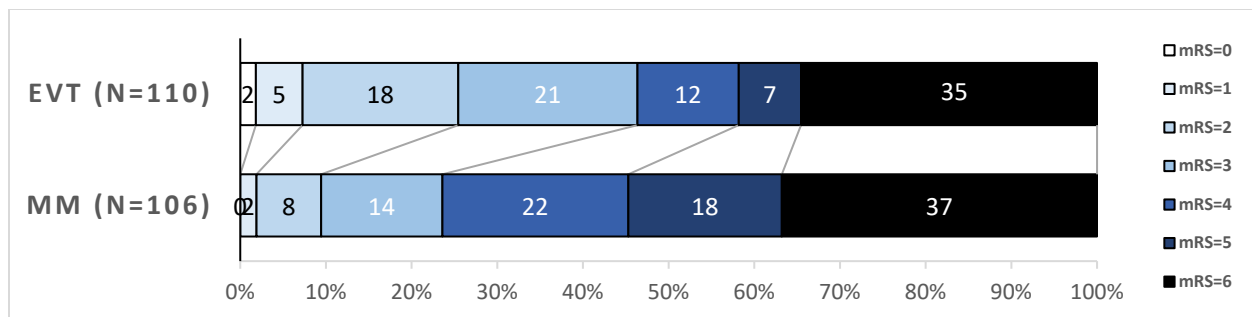
ac) in patients with mismatch ratio ≥ 1.2 & mismatch volume $\geq 10\text{ml}$ based on composite core:



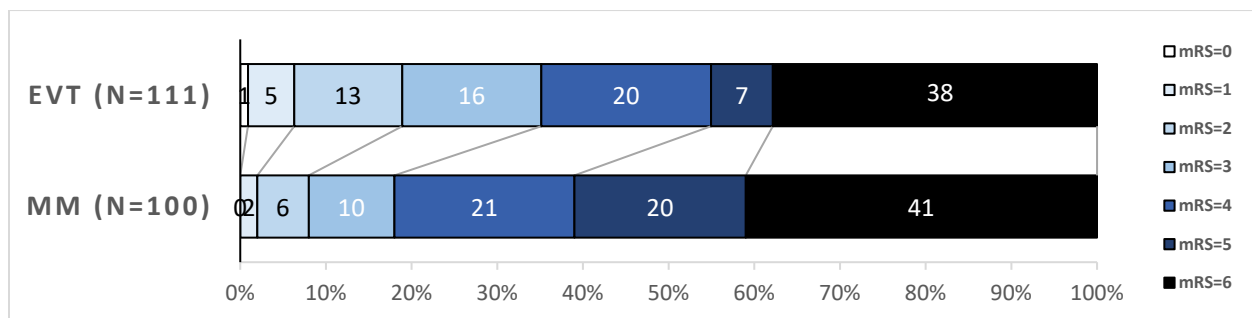
ad) in patients without mismatch ratio ≥ 1.8 & mismatch volume $\geq 15\text{ml}$ based on CTP/MRI core:



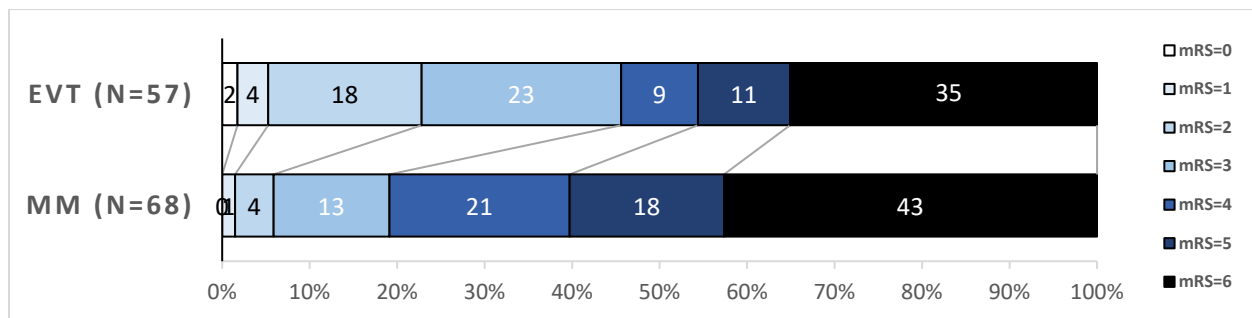
ae) in patients with mismatch ratio ≥ 1.8 & mismatch volume $\geq 15\text{ml}$ based on CTP/MRI core:



af) in patients without mismatch ratio ≥ 1.8 & mismatch volume ≥ 15 ml based on composite core:



ag) in patients with mismatch ratio ≥ 1.8 & mismatch volume ≥ 15 ml based on composite core:



eTables

eTable 1. Baseline clinical and imaging characteristics of study population (as-treated), stratified based on treatment.

		Endovascular thrombectomy	Medical care only
		N=170	N=166
Demographics			
Age (years)		66.0 (59.0-75.0)	67.0 (58.0-75.0)
Sex	Female	68 (40.0%)	71 (42.8%)
	Male	102 (60.0%)	95 (57.2%)
Race	Asian	5 (2.9%)	3 (1.8%)
	Black	24 (14.1%)	24 (14.5%)
	Caucasian	126 (74.1%)	123 (74.1%)
	Native Hawaiian or Pacific Islander	2 (1.2%)	0
	Other or unknown	13 (7.6%)	16 (9.6%)
Transferred to thrombectomy-Capable Center	Directly Presenting to thrombectomy-capable Center	71 (41.8%)	65 (39.2%)
	Transferred to thrombectomy-Capable Center	99 (58.2%)	101 (60.8%)
Medical History			
H/o Hypertension		132 (77.6%)	120 (72.3%)
H/o Diabetes Mellitus		50 (29.4%)	54 (32.5%)
H/o Atrial Fibrillation		44 (25.9%)	38 (22.9%)
H/o Coronary Artery Disease		39 (23.2%)	25 (15.6%)
H/o Congestive Heart Failure		20 (11.8%)	18 (10.8%)
H/o Ischemic Stroke		18 (10.6%)	13 (7.8%)
H/o Transient Ischemic Attack		4 (2.4%)	8 (4.8%)
Physical Examination			
Hemisphere affected		78 (45.9%)	69 (41.6%)
NIH Stroke Scale at thrombectomy hospital		19 (15-23)	19 (15-22)
Neuroimaging findings			
Occlusion location	ICA	75 (44.1%)	64 (38.6%)
	MCA M1	88 (51.8%)	94 (56.6%)
	MCA M2	7 (4.1%)	8 (4.8%)
Tandem occlusions		54 (31.8%)	43 (25.9%)

	Endovascular thrombectomy	Medical care only
CT ASPECTS at thrombectomy hospital	4 (3-5)	4 (4-5)
Composite core volume (ml)	103 (69-140)	100 (74-136)
CTP/MRI core volume (ml)	70 (39-110)	77 (48-104)
CT Hypodensity volume (ml)	84 (45-114)	87 (49-113)
Critically hypoperfused (Tmax>6s) volume (ml)	161 (117-206)	166 (119-212)
Time metrics		
Time from Last Known Well to Randomization (minutes)	550 (311-920)	588 (345-928)
Time from Arrival to CT Acquisition (minutes)	16 (9-27)	16 (7-24)
Time from Arrival to CTP Acquisition (minutes)	26 (17-42)	25 (13-36)
Additional characteristics		
IV Thrombolytics Administered	33 (19.4%)	28 (17.0%)
Tenecteplase used	4 (12.5%)	1 (3.7%)
General Anesthesia Used	100 (59.5%)	

Footnotes:

CT – computed tomography

MRI – Magnetic Resonance Imaging

NIH Stroke Scale – a 42 point scoring system measuring neurological deficits, with higher score representing more neurological deficits.

Tmax – Time to maximum intensity

CT Alberta Stroke Program Early CT Score (ASPECTS) – a 10 point ordinal scoring system measuring areas of hypoattenuation due to ischemic changes in the middle cerebral artery territory on non-contrast CT of brain, with a point deducted for each location demonstrating hypoattenuation. Thus, lower scores represent worse the ischemic injury.

CTP/MRI core volume – volumetric estimates obtained from CT perfusion using relative Cerebral Blood Flow <30% threshold/MR perfusion imaging using apparent diffusion coefficient <620 X10⁻⁶ mm/s² threshold using automated RAPIDai processing.

CT hypodensity volume – volumetric estimates of observed hypodensity on non-contrast CT by manual delineation of the region of interest by consensus from 3 expert readers using ITK-SNAP
Composite core volume – the larger of CTP/MRI core volume and CT hypodensity volume estimates.

eTable 2. Clinical and imaging outcomes in intention-to-treat population, stratified by treatment.

	Endovascular Thrombectomy Medical Care	Plus Medical Care Only
	N=168	N=168
Distribution of Modified Rankin Scale Scores at 90d Follow-up	4 (3-6)	5 (4-6)
Functional independence (mRS 0-2) at 90-day follow-up	34/168 (20.2%)	12/168 (7.1%)
Independent Ambulation (mRS 0-3) at 90-day follow-up	65/168 (38.7%)	31/168 (18.5%)
Complete dependence or death (mRS 5-6) at 90d follow-up	76/168 (45.2%)	102/168 (60.7%)
All-cause Mortality within 90-day Follow-up Period	62/168 (36.9%)	70/168 (41.7%)
Symptomatic Intracerebral hemorrhage	0/168 (0.0%)	2/168 (1.2%)
Parenchymal hemorrhage	4/168 (2.4%)	3/168 (1.8%)
Neurological Worsening	38/168 (22.6%)	27/168 (16.1%)
Early Neurological Improvement	19/165 (11.5%)	13/166 (7.8%)
Follow-up infarct volume	170 (123-266)	168 (109-254)
Infarct growth from CTP/MRI core (ml)	100 (42-177)	90 (53-140)
Infarct growth from composite core (ml)	68 (22-128)	66 (24-110)
Infarct growth from CT hypodensity (ml)	87 (37-152)	73 (31-135)

eTable 3. Sensitivity analyses using site-investigator adjudicated ASPECTS.

ASPECTS category	mRS (aGenOR 95% CI) shift with	mRS 0-2 (aRR with 95% CI)	mRS 0-3 (aRR with 95% CI)	mRS 5-6 (aRR with 95% CI)
ASPECTS 3-5	aGenOR: 1.70, 95% CI: 1.33-2.17	aRR: 2.97, 95% CI: 1.60-5.49	aRR: 2.23, 95% CI: 1.56-3.16	aRR: 0.74, 95% CI: 0.61-0.91
ASPECTS 6-10	aGenOR: 1.91, 95% CI: 1.04-3.54	aRR: 2.93, 95% CI: 0.42-20.47	aRR: 1.91, 95% CI: 0.75-4.90	aRR: 0.60, 95% CI: 0.35-1.02
ASPECTS 3	aGenOR: 1.88, 95% CI: 1.27-2.78	aRR: 7.82, 95% CI: 1.14-53.56	aRR: 2.53, 95% CI: 1.00-6.36	aRR: 0.66, 95% CI: 0.49-0.88
ASPECTS 4	aGenOR: 1.51, 95% CI: 0.92-2.48	aRR: 4.11, 95% CI: 1.40-12.09	aRR: 2.44, 95% CI: 1.39-4.29	aRR: 0.94, 95% CI: 0.62-1.44
ASPECTS 5	aGenOR: 1.93, 95% CI: 1.25-2.98	aRR: 2.08, 95% CI: 1.02-4.22	aRR: 2.46, 95% CI: 1.56-3.90	aRR: 0.71, 95% CI: 0.50-1.01

eTable 4. Sensitivity analyses evaluating EVT treatment effect across CTP core strata (restricted to patients who received CT Perfusion only).

CTP core (rCBF <30%) volume categories	mRS shift (aGenOR with 95% CI)	mRS 0-2 (aRR with 95% CI)	mRS 0-3 (aRR with 95% CI)	mRS 5-6 (aRR with 95% CI)
<70ml	aGenOR: 1.73, 95% CI: 1.20-2.48	aRR: 2.26, 95% CI: 1.15-4.43	aRR: 1.76, 95% CI: 1.24-2.50	aRR: 0.77, 95% CI: 0.53-1.11
≥70ml	aGenOR: 1.61, 95% CI: 1.21-2.15	aRR: 4.12, 95% CI: 1.25-13.63	aRR: 2.89, 95% CI: 1.39-6.02	aRR: 0.75, 95% CI: 0.61-0.92
<100ml	aGenOR: 1.86, 95% CI: 1.40-2.49	aRR: 2.75, 95% CI: 1.49-5.07	aRR: 2.05, 95% CI: 1.45-2.89	aRR: 0.68, 95% CI: 0.52-0.88
≥100ml	aGenOR: 1.44, 95% CI: 1.00-2.06	aRR: 3.36, 95% CI: 0.52-21.64	aRR: 3.04, 95% CI: 0.95-9.73	aRR: 0.82, 95% CI: 0.64-1.04
<150ml	aGenOR: 1.78, 95% CI: 1.39-2.30	aRR: 2.93, 95% CI: 1.62-5.30	aRR: 2.10, 95% CI: 1.51-2.92	aRR: 0.70, 95% CI: 0.56-0.87
≥150ml	aGenOR: 1.43, 95% CI: 0.80-2.54	N/A	N/A	aRR: 0.83, 95% CI: 0.56-1.23

N/A – could not be evaluated

eTable 5. Association of extent of ischemic injury on CT hypodensity and composite core volumes and clinical outcomes within patients receiving EVT and MM.

	Endovascular Thrombectomy (N=170)	Medical Care Only (N=166)	Interaction p-value
mRS Distribution	Median (IQR): 4 (3-6)	Median (IQR): 5 (4-6)	
CT hypodensity volume(per 10ml increment)	aGenOR: 0.95, 95% CI: 0.93-0.98	aGenOR: 0.98, 95% CI: 0.95-1.00	0.14
Composite core volume(per 10ml increment)	aGenOR: 0.92, 95% CI: 0.88-0.95	aGenOR: 0.95, 95% CI: 0.92-0.98	0.13
mRS 0-2	34/170 (20.0%)	12/166 (7.2%)	
CT hypodensity volume(per 10ml increment)	aRR: 0.93, 95% CI: 0.89- 0.98	aRR: 0.97, 95% CI: 0.87-1.09	0.53
Composite core volume(per 10ml increment)	aRR: 0.87, 95% CI: 0.82- 0.93	aRR: 0.93, 95% CI: 0.83-1.05	0.82
mRS 0-3	66/170 (38.8%)	30/166 (18.1%)	
CT hypodensity volume(per 10ml increment)	aRR: 0.97, 95% CI: 0.91- 1.03	aRR: 0.97, 95% CI: 0.91-1.03	0.32
Composite core volume(per 10ml increment)	aRR: 0.91, 95% CI: 0.85- 0.98	aRR: 0.91, 95% CI: 0.85-0.98	0.82
mRS 5-6	77/170 (45.3%)	101/166 (60.8%)	
CT hypodensity volume(per 10ml increment)	aRR: 1.04, 95% CI: 1.02- 1.07	aRR: 1.03, 95% CI: 1.01-1.05	0.76
Composite core volume(per 10ml increment)	aRR: 1.03, 95% CI: 1.00- 1.05	aRR: 1.01, 95% CI: 1.00-1.03	0.63

Footnotes:

CT- computed tomography

CTP – CT perfusion

MRI – Magnetic resonance imaging

ASPECTS – Alberta Stroke Program Early CT Score, a 10 point ordinal scoring system measuring areas of hypoattenuation due to ischemic changes in the middle cerebral artery territory on non-contrast CT of brain, with a point deducted for each location demonstrating hypoattenuation. Thus, lower scores represent worse ischemic injury.

mRS – modified Rankin Scale score, a 7 level ordinal scoring system representing functional status, with higher scores representing worse functional status (0 being without any deficits and 6 being dead).

eTable 6. Evaluation parameters for regression models for functional outcomes

Outcome	Modality	AIC	BIC	AUC (with 95% CI)	Preferred modality (based on BIC conventions)
Ordinal mRS	ASPECTS	417.2590	447.8850	NA	
	CT hypodensity volume	412.6195	443.2454	NA	
	CTP/MRI core volume	403.2944	433.9203	NA	
	Composite core volume	398.6922	429.3182	NA	Preferred
mRS 0-2	ASPECTS	256.9212	276.0068	0.7106 (0.6343-0.7870)	
	CT hypodensity volume	252.1431	271.2287	0.7224 (0.6481-0.7968)	
	CTP/MRI core volume	239.4032	258.4887	0.7716 (0.7062-0.8370)	
	Composite core volume	236.5431	255.6287	0.7783 (0.7103-0.8463)	Preferred
mRS 0-3	ASPECTS	358.6683	377.7538	0.7500 (0.6937-0.8062)	
	CT hypodensity volume	352.1023	371.1878	0.7522 (0.6965-0.8079)	
	CTP/MRI core volume	329.0319	348.1174	0.7952 (0.7454-0.8450)	
	Composite core volume	324.2519	343.3374	0.7977 (0.7472-0.8482)	Preferred

Parameters for mRS distribution were calculated using ordinal logistic regression.

Models with lower AIC and BIC values are considered more parsimonious

Models with higher AUC values are considered to have better predictive value.

N/A – not applicable

eTable 7. Association of presence or absence of mismatch profile based on composite core volume with clinical outcomes within patients receiving EVT and MM (as-treated analysis).

Presence of mismatch profile (mismatch volume ≥ 10 ml & mismatch ratio ≥ 1.2) – based on Composite core volume	mRS distribution			
	No mismatch (ref)	4 (3, 6)	5 (4, 6)	
	Mismatch	4 (3, 6)	5 (4, 6)	
	Effect size with uncertainty estimates	aGenOR: 0.77, 95% CI: 0.50-1.17	aGenOR: 0.69, 95% CI: 0.46-1.05	0.73
	Functional independence (mRS 0-2)			
	No mismatch (ref)	5/34 (14.7%)	3/40 (7.5%)	
	Mismatch	29/136 (21.3%)	9/126 (7.1%)	
	Effect size with uncertainty estimates	aRR: 1.02, 95% CI: 0.48-2.15, aRD:-0.000, 95% CI: -0.155 to 0.154	aRR: 0.63, 95% CI: 0.20-1.94, aRD:-0.034, 95% CI: -0.123 to 0.056	0.47
	Independent ambulation (mRS 0-3)			
	No mismatch (ref)	13/34 (38.2%)	7/40 (17.5%)	
	Mismatch	53/136 (39.0%)	23/126 (18.3%)	
	Effect size with uncertainty estimates	aRR: 0.77, 95% CI: 0.48-1.26, aRD:-0.097, 95% CI: -0.263 to 0.070	aRR: 0.68, 95% CI: 0.33-1.38, aRD:-0.064, 95% CI: -0.184 to 0.055	0.79
	Complete dependence or death (mRS 5-6)			
	No mismatch (ref)	12/34 (35.3%)	23/40 (57.5%)	
	Mismatch	65/136 (47.8%)	78/126 (61.9%)	
	Effect size with uncertainty estimates	aRR: 1.31, 95% CI: 0.80-2.15, aRD:0.127, 95% CI: -0.039 to 0.293	aRR: 1.29, 95% CI: 0.93-1.79, aRD:0.168, 95% CI: 0.011 to 0.325	0.83
Presence of mismatch profile (mismatch volume ≥ 15 ml)	mRS distribution			
	No mismatch (ref)	4 (3, 6)	5 (4, 6)	
	Mismatch	4 (3, 6)	5 (4, 6)	

& mismatch ratio ≥ 1.8) – based on Composite core volume	Effect size with uncertainty estimates	aGenOR: 0.76, 95% CI: 0.50-1.18	aGenOR: 0.73, 95% CI: 0.53-1.02	0.88
	Functional independence (mRS 0-2)			
	No mismatch (ref)	21/112 (18.8%)	8/99 (8.1%)	
	Mismatch	13/58 (22.4%)	4/67 (6.0%)	
	Effect size with uncertainty estimates	aRR: 0.60, 95% CI: 0.30-1.21, aRD: -0.099, 95% CI: -0.224 to 0.027	aRR: 0.44, 95% CI: 0.15-1.26, aRD: -0.061, 95% CI: -0.144 to 0.022	0.76
	Independent ambulation (mRS 0-3)			
	No mismatch (ref)	39/112 (34.8%)	18/99 (18.2%)	
	Mismatch	27/58 (46.6%)	12/67 (17.9%)	
	Effect size with uncertainty estimates	aRR: 0.81, 95% CI: 0.55-1.21, aRD: -0.076, 95% CI: -0.228 to 0.075	aRR: 0.66, 95% CI: 0.38-1.14, aRD: -0.091, 95% CI: -0.202 to 0.020	0.68
	Complete dependence or death (mRS 5-6)			
	No mismatch (ref)	51/112 (45.5%)	60/99 (60.6%)	
	Mismatch	26/58 (44.8%)	41/67 (61.2%)	
	Effect size with uncertainty estimates	aRR: 1.09, 95% CI: 0.74-1.61, aRD: 0.068, 95% CI: -0.088 to 0.224	aRR: 1.21, 95% CI: 0.94-1.56, aRD: 0.121, 95% CI: -0.017 to 0.260	0.79

mRS – modified Rankin Scale score, a 7 level ordinal scoring system representing functional status, with higher scores representing worse functional status (0 being without any deficits and 6 being dead).

Analyses are adjusted for age, NIHSS, time from last known well to randomization and CTP/MRI core volume.

eTable 8. Association of concordance/discordance between CT ASPECTS and ischemic core estimates on CTP/MRI, within patients receiving EVT and MM.

	Endovascular Thrombectomy (N=163)		Medical Care Only (N=156)		Interaction p-value
mRS Distribution	Median (IQR): 4 (3-6)	Effect size with 95% CI	Median (IQR): 5 (4-6)	Effect size with 95% CI	
Concordant imaging profiles (ASPECTS<6 & CTP/MRI core≥70ml)	5 (4-6)	Ref	5 (4-6)	Ref	
Discordant imaging profile – 1(ASPECTS<6 & CTP/MRI core<70ml)	3 (2-6)	aGenOR: 0.91 (0.50, 1.65)	4 (3-6)	aGenOR: 1.50, 95% CI: 0.94-2.40	0.20
Discordant imaging profile – 2(ASPECTS 6-10 & CTP/MRI core≥70ml)	5 (3.5-6)	aGenOR: 1.01 (0.53-1.92)	6 (6-6)	aGenOR: 0.65, 95% CI: 0.43-0.99	0.26
mRS 0-2	N (%): 32/163 (19.6%)	Effect size with 95% CI	N (%): 10/156 (6.4%)	Effect size with 95% CI	
Concordant imaging profiles(ASPECTS<6 & CTP/MRI core≥70ml)	10/75 (13%)	Ref	3/82 (4%)	Ref	
Discordant imaging profile – 1(ASPECTS<6 & CTP/MRI core<70ml)	21/76 (28%)	aRR: 0.51 (0.16-1.56) aRD: -0.055 (-0.201, 0.092)	7/63 (11%)	aRR: 1.26 (0.17-9.43) aRD: 0.040 (-0.059, 0.138)	0.78
Discordant imaging profile – 2(ASPECTS 6-10 & CTP/MRI core≥70ml)	1/12 (8%)	aRR: 0.85 (0.12-5.79) aRD: 0.010 (-0.395, 0.415)	0/11 (0%)	N/A	N/A
mRS 0-3	N (%): 62/163 (38.0%)	Effect size with 95% CI	N (%): 27/156 (17.3%)	Effect size with 95% CI	
Concordant imaging profiles(ASPECTS<6 & CTP/MRI core≥70ml)	18/75 (24%)	Ref	8/82 (10%)	Ref	
Discordant imaging profile –	41/76 (54%)	aRR: 0.86 (0.43-1.71)	19/63 (30%)	aRR: 1.24 (0.38-4.07)	0.72

1(ASPECTS<6 & CTP/MRI core<70ml)		aRD: 0.043 (-0.125, 0.211)		aRD: 0.060 (-0.075, 0.196)	
Discordant imaging profile – 2(ASPECTS 6-10 & CTP/MRI core≥70ml)	3/12 (25%)	aRR: 1.19 (0.43-3.28) aRD: -0.065 (-0.421, 0.291)	0/11 (0%)	N/A	N/A
mRS 5-6	N (%): 75/163 (46.0%)	Effect size with 95% CI	N (%): 97/156 (62.2%)	Effect size with 95% CI	
Concordant imaging profiles(ASPECTS<6 & CTP/MRI core≥70ml)	42/75 (56%)	Ref	59/82 (72%)	Ref	
Discordant imaging profile – 1(ASPECTS<6 & CTP/MRI core<70ml)	25/76 (33%)	aRR: 0.92 (0.57-1.48) aRD: -0.077 (-0.244, 0.091)	28/63 (44%)	aRR: 0.69 (0.49-0.97) aRD: -0.076 (-0.248, 0.096)	0.14
Discordant imaging profile – 2(ASPECTS 6-10 & CTP/MRI core≥70ml)	8/12 (67%)	aRR: 1.08 (0.69-1.68) aRD: 0.143 (-0.212, 0.498)	10/11 (91%)	aRR: 1.28 (1.01-1.63) aRD: 0.324 (0.165, 0.483)	0.02

Footnotes:

CT- computed tomography

CTP – CT perfusion

MRI – Magnetic resonance imaging

ASPECTS – Alberta Stroke Program Early CT Score, a 10 point ordinal scoring system measuring areas of hypoattenuation due to ischemic changes in the middle cerebral artery territory on non-contrast CT of brain, with a point deducted for each location demonstrating hypoattenuation. Thus, lower scores represent worse ischemic injury.

mRS – modified Rankin Scale score, a 7 level ordinal scoring system representing functional status, with higher scores representing worse functional status (0 being without any deficits and 6 being dead).

eTable 9. Estimates of EVT treatment effect and corresponding interaction terms for concordant and discordant imaging profiles based on CTP/MRI core volume estimates

	mRS distribution			
	MM	EVT	mRS distribution(aGenOR with 95% CI)	Interaction p-value
Concordant imaging profiles(ASPECTS<6 & CTP/MRI core≥70ml)	5 (4, 6)	5 (4, 6)	aGenOR: 1.60, 95% CI: 1.18 to 2.18	Ref
Discordant imaging profile – 1(ASPECTS<6 & CTP/MRI core<70ml)	4 (3, 6)	3 (2, 6)	aGenOR: 1.87, 95% CI: 1.28 to 2.74	0.66
Discordant imaging profile – 2(ASPECTS 6-10 & CTP/MRI core≥70ml)	6 (6, 6)	5 (3.5, 6)	aGenOR: 2.46, 95% CI: 1.14 to 5.32	0.18
	Functional independence(mRS 0-2)			
	MM	EVT	aRR with 95% CI	Interaction p-value
Concordant imaging profiles(ASPECTS<6 & CTP/MRI core≥70ml)	3/83 (3.6%)	10/74 (13.5%)	aRR: 4.62, 95% CI: 1.27 to 16.71	Ref
Discordant imaging profile – 1(ASPECTS<6 & CTP/MRI core<70ml)	7/63 (11.1%)	21/76 (27.6%)	aRR: 2.68, 95% CI: 1.25 to 5.76	0.56
Discordant imaging profile – 2(ASPECTS 6-10 & CTP/MRI core≥70ml)	1/11 (9.1%)	0/12 (0%)	N/A	N/A
	Independent ambulation(mRS 0-3)			
	MM	EVT	aRR with 95% CI	Interaction p-value
Concordant imaging profiles(ASPECTS<6 & CTP/MRI core≥70ml)	8/83 (9.6%)	18/74 (24.3%)	aRR: 2.98, 95% CI: 1.44 to 6.17	Ref
Discordant imaging profile – 1(ASPECTS<6 & CTP/MRI core<70ml)	19/63 (30.2%)	41/76 (53.9%)	aRR: 1.90, 95% CI: 1.29 to 2.78	0.38
Discordant imaging profile – 2(ASPECTS 6-10 & CTP/MRI core≥70ml)	3/11 (27.3%)	0/12 (0%)	N/A	N/A
	Complete dependence or death (mRS 5-6)			
	MM	EVT	aRR with 95% CI	Interaction p-value
Concordant imaging profiles(ASPECTS<6 & CTP/MRI core≥70ml)	60/83 (72.3%)	41/74 (55.4%)	aRR: 0.76, 95% CI: 0.61 to 0.95	Ref

Discordant imaging profile – 1(ASPECTS<6 & CTP/MRI core<70ml)	28/63 (44.4%)	25/76 (32.9%)	aRR: 0.72, 95% CI: 0.49 to 1.05	0.83
Discordant imaging profile – 2(ASPECTS 6-10 & CTP/MRI core≥70ml)	10/11 (90.9%)	8/12 (66.7%)	aRR: 0.59, 95% CI: 0.37 to 0.96	0.40

N/A – could not be evaluated

eTable 10. Estimates of EVT treatment effect and corresponding interaction terms for concordant and discordant imaging profiles based on composite core volume estimates

	mRS distribution			
	MM	EVT	mRS distribution(aGenOR with 95% CI)	Interaction p-value
Concordant imaging profiles(ASPECTS<6 & composite core≥70ml)	5 (4, 6)	4 (3, 6)	aGenOR: 1.65, 95% CI: 1.26 to 2.16	Ref
Discordant imaging profile – 1(ASPECTS<6 & composite core<70ml)	4.5 (3, 6)	3 (2, 6)	aGenOR: 1.80, 95% CI: 1.00 to 3.22	0.61
Discordant imaging profile – 2(ASPECTS 6-10 & composite core≥70ml)	6 (6, 6)	5 (3, 6)	aGenOR: 2.66, 95% CI: 1.34 to 5.28	0.13
	Functional independence(mRS 0-2)			
	MM	EVT	aRR with 95% CI	Interaction p-value
Concordant imaging profiles(ASPECTS<6 & composite core≥70ml)	8/118 (6.8%)	18/112 (16.1%)	aRR: 2.88, 95% CI: 1.34 to 6.20	Ref
Discordant imaging profile – 1(ASPECTS<6 & composite core<70ml)	2/28 (7.1%)	13/38 (34.2%)	aRR: 3.71, 95% CI: 0.93 to 14.82	0.71
Discordant imaging profile – 2(ASPECTS 6-10 & composite core≥70ml)	2/12 (16.7%)	0/14 (0%)	N/A	N/A
	Independent ambulation(mRS 0-3)			
	MM	EVT	aRR with 95% CI	Interaction p-value
Concordant imaging profiles(ASPECTS<6 & composite core≥70ml)	19/118 (16.1%)	35/112 (31.3%)	aRR: 2.21, 95% CI: 1.41 to 3.48	Ref
Discordant imaging profile – 1(ASPECTS<6 & composite core<70ml)	8/28 (28.6%)	24/38 (63.2%)	aRR: 1.83, 95% CI: 1.04 to 3.22	0.68
Discordant imaging profile – 2(ASPECTS 6-10 & composite core≥70ml)	4/12 (33.3%)	0/14 (0%)	N/A	N/A
	Complete dependence or death (mRS 5-6)			
	MM	EVT	aRR with 95% CI	Interaction p-value
Concordant imaging profiles(ASPECTS<6 & composite core≥70ml)	74/118 (62.7%)	53/112 (47.3%)	aRR: 0.75, 95% CI: 0.60 to 0.92	Ref

Discordant imaging profile – 1(ASPECTS<6 & composite core<70ml)	14/28 (50.0%)	13/38 (34.2%)	aRR: 0.90, 95% CI: 0.52 to 1.56	0.92
Discordant imaging profile – 2(ASPECTS 6-10 & composite core≥70ml)	11/12 (91.7%)	9/14 (64.3%)	aRR: 0.59, 95% CI: 0.39 to 0.88	0.38

N/A – could not be evaluated

eTable 11. Estimates of infarct growth from baseline in patients with MR DWI follow-up, stratified based on treatment arm and reperfusion status.

Factor	MM	mTICI 0-2a	mTICI 2b-3
N	101	24	79
Infarct growth from CTP/MRI core (ml), median (IQR)	95 (56, 135)	125 (76, 179)	68 (37, 142)
Infarct growth from composite core (ml), median (IQR)	63 (22, 102)	91 (71, 142)	45 (16, 90)
Infarct growth from CT hypodensity (ml), median (IQR)	69 (31, 122)	107 (87, 152)	60 (24, 110)

eTable 12. Comparison of infarct growth in study population, based on functional outcomes.

	Functionally Dependent (mRS 3-6)	Functionally Independent (mRS 0-2)	p-value
	N=290	N=46	
Infarct growth from CTP/MRI core (ml)	102 (61-167)	45 (32-88)	<0.001
Infarct growth from composite core (ml)	77 (31-127)	23.5 (11-48)	<0.001
	No independent ambulation (mRS 4-6)	Independent Ambulation (mRS 0-3)	p-value
	N=240	N=96	
Infarct growth from CTP/MRI core (ml)	110.5 (65.5-174.5)	65.5 (32.5-100.5)	<0.001
Infarct growth from composite core (ml)	85.5 (36-144)	31.5 (10-67)	<0.001

eTable 13. Characteristics of patients demonstrating significant(≥ 10 ml) overestimation of ischemic core as compared to follow-up infarct volume.

ID	CTP or MRI core volume(ml)	CT hypodensity volume(ml)	Follow-up infarct volume(ml)	Time from last known well to CTP acquisition(minutes)	Time from CTP acquisition to reperfusion/end of the procedure(min)
1	162	33	143	642	77
2	140	0	90	289	N/A
3	93	5	11	51	125

N/A – not applicable (patient received medical management)